AN EXOTIC $S^2 \times S^2$ AND AN EXOTIC $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$

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ABSTRACT. We prove that a specified Lidman–Piccirillo piece V, a symplectic 4-manifold with the homology of $S^2 \times D^2$ built from a genus-2 surface bundle over a once-punctured torus by two Luttinger surgeries, is simply connected, for an explicit permitted choice of the two surgery parametrizations. Three consequences follow. The symplectic double $Z = V \cup_\sigma V$ is homeomorphic but not diffeomorphic to $S^2 \times S^2$. The Lidman–Piccirillo manifolds B and W are homeomorphic. Since the figure-eight knot is slice in B and not in W, they are the first pair of homeomorphic closed 4-manifolds distinguished by unconstrained knot slicing, that is, by sliceness with no constraint on the homology class of the slice disk; detecting smooth structure this way goes back to Casson. Finally, the regluing of Lidman and Piccirillo's Theorem 2 applied to Z yields a closed simply connected 4-manifold homeomorphic but not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. The consequences follow from the simple-connectivity statement by the classifications of Freedman and of Hambleton–Kreck, together with a rigidity analysis of the surgery parameters. The fundamental group is computed in the style of Baldridge and Kirk, from explicit based representatives of every meridian and Lagrangian push off, and the resulting relation system is decided by coset enumeration, after calibration on two configurations whose answers are known independently. The development calculations and finite-presentation decisions reproduce from the ancillary files.

1. Introduction

One approach to distinguishing smooth structures on 4-manifolds, going back to Casson, is to find a knot that is smoothly slice in one manifold and not in another with the same underlying topology. Lidman and Piccirillo [LP25] recently carried out the first successful version of this program at the level of cohomology rings:

Theorem 1.1 ([LP25, Theorem 1]). *There are spin rational homology four-spheres B and W with $H_1 = \mathbb{Z}/2$ and isomorphic integer cohomology rings such that the figure-eight knot 4_1 is slice in B but not in W.*

Here B is the Kawauchi manifold [Kaw09] and $W = V/\sigma$, where V is a symplectic homology $S^2 \times D^2$ built from a genus-2 surface bundle R over a once-punctured torus by two Luttinger surgeries, and σ is a free involution of $\partial V = S^3_0(Q)$, Q the square knot. This paper proves the three theorems below.

Theorem A (Exotic $S^2 \times S^2$). *The symplectic double $Z = V \cup_\sigma V$ is homeomorphic to $S^2 \times S^2$ and not diffeomorphic to it.*

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Theorem B (A homeomorphic pair distinguished by slicing)**.** *W is homeomorphic to B. Since the figure-eight knot is slice in B and not slice in W, the pair (B, W) consists of homeomorphic, non-diffeomorphic closed 4-manifolds distinguished by unconstrained knot slicing.*[1]

Theorem C (Exotic $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$)**.** *The regluing of* [LP25, Theorem 2] *applied to Z is a closed simply connected 4-manifold homeomorphic to $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$ and not diffeomorphic to it.*

An exotic $S^2 \times S^2$, or an exotic $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$, is a long-standing open problem; the only claim in the literature, by Akhmedov and Park [AP10], has remained an unpublished preprint since 2010 and is cited as such by the current literature [SS23, BSS24], while [LP25] re-proves the strictly weaker "nonstandard cohomology $S^2 \times S^2$" statement (their Theorem 8) and describes the known examples as "presumably not simply connected". The smallest closed simply connected 4-manifolds with signature zero known to admit exotic smooth structures are the connected sums $\#_{2m+1}(\mathbb{CP}^2\#\overline{\mathbb{CP}}^2)$ for $m \geq 4$ and $\#_{2n+1}(S^2 \times S^2)$ for $n \geq 5$ of Baykur–Hamada [BH23], who obtain smaller topology only at the price of a nontrivial fundamental group; Theorems A and C concern the bottom case of each family. Theorem B is the upgrade of Theorem 1.1 from isomorphic cohomology rings to homeomorphism, and is the first pair of homeomorphic closed 4-manifolds distinguished by unconstrained slicing.

Lidman and Piccirillo explicitly leave the parametrizing curves on the two surgery tori arbitrary, so their notation V does not specify a unique manifold before those choices are made. Throughout this paper, V means the result of their two $+1$ Luttinger surgeries when the two base-direction curves are the Lagrangian push offs developed in Sections 8.4–8.5; in the notation of Proposition 1.3, this is $V = V'_{0,0}$. The plus signs here refer to the geometric surgery convention of [LP25]; their exponents in the based relator sheet depend on orientation choices and are included among the signs of Convention 2.1. This defines the object considered here; it is not a claim that the parametrization is canonical.

All three theorems come from a single group-theoretic fact.

Theorem D (Triviality of π_1)**.** *For the specified Lidman–Piccirillo piece $V = V'_{0,0}$, one has $\pi_1(V) = 1$.*

For the fixed piece, the geometrically derived relation system lies in a finite robustness family that independently varies the five signs in the relator sheet, the left/right placement of all three transport corrections, both conjugating arcs for the Bs correction, and both arc/sign/placement choices in the T_β push off. Coset enumeration terminates with order one in all 4,096 cases. Knuth–Bendix computations on nearby formal exponent choices and a second based diagram are retained in Appendix A.2 as diagnostics; no simple-connectivity claim for those additional parametrizations is needed or made.

The two parts. Part I (Sections 3–6) proves that Theorem D implies Theorems A–C, and that the implication is in an appropriate sense an equivalence. Call a 4-manifold V' an *admissible variant* of V if it is obtained from R by Luttinger surgeries from the family specified in Proposition 1.3 below: one surgery on each of T_α, T_β, with the listed coefficients and directions. This is a definition of the family, not a classification of all Luttinger surgeries that could produce the same homology. The fixed Lidman–Piccirillo choice is the case $(m, n) = (0, 0)$. Set $W' = V'/\sigma$ and $Z' = V' \cup_\sigma V'$.

[1] *Unconstrained*: no constraint is placed on the homology class of the slice disk.

Theorem 1.2 (Reduction)**.** *Let V' be any admissible variant. Then:*

(a) $W' \cong_{\text{homeo}} B$ *if and only if* $\pi_1(Z') = 1$.
(b) *If* $\pi_1(Z') = 1$, *then* Z' *is homeomorphic to* $S^2 \times S^2$ *and not diffeomorphic to it, and the pair* (B, W') *is homeomorphic and non-diffeomorphic, distinguished by the sliceness of* 4_1.
(c) *If* $\pi_1(V') = 1$ *(which implies* $\pi_1(Z') = 1$*), then in addition the regluing of* [LP25, Theorem 2] *applied to* Z' *yields a simply connected* 4*-manifold homeomorphic but not diffeomorphic to* $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.

Theorem 1.2 runs in both directions: the slicing upgrade cannot be obtained without solving the exotic $S^2 \times S^2$ problem inside a σ-symmetric double, and conversely any solution of the simple-connectivity problem in the Lidman–Piccirillo setting yields three theorems simultaneously (strictly more than the Akhmedov–Park setting, which yields the exotic $S^2 \times S^2$ alone), with no additional work, since surgeries in $\operatorname{int} V$ descend to the quotient automatically. This, we suggest, is the difficulty at which [LP25] stops.

The proof of Theorem 1.2(a) rests on the topological classification of 4-manifolds with finite cyclic fundamental group: by Freedman and Hambleton–Kreck [Fre82, HK88, HK93] (see [Ham08, Theorem 5.1] for the statement in this form, and [BSS24] for its $b_2 = 0$ case), a closed oriented topological 4-manifold with $\pi_1 = \mathbb{Z}_n$ is determined up to homeomorphism by its intersection form, w_2-type, and Kirby–Siebenmann invariant; in particular the smooth spin rational homology 4-sphere with $\pi_1 = \mathbb{Z}/2$ is unique up to homeomorphism. The input on the smooth side is a rigidity statement for the surgery parameters:

Proposition 1.3 (Rigidity of the surgery parameters)**.** *If one Luttinger surgery is performed on each of* $T_\alpha, T_\beta \subset R$ *and the result is a homology* $S^2 \times D^2$, *then both surgeries have coefficient* ± 1, *and their directions equal* α', β' *modulo fiber classes (throughout,* α', β' *denote curves in* R *projecting to the base curves* α, β*; their classes generate* $H_1(R)$, *Lemma 4.2). Conversely, for every* $m, n \in \mathbb{Z}$ *the directions* $\beta' e^m, \alpha' c^n$ *with coefficients* ± 1 *yield* $H_1(V') = 0$ *and preserve: the spin condition, symplecticity,* $\chi = 2$, $H_2(V') = \mathbb{Z}\langle F \rangle$, *the descent of* σ, *and the hypotheses of the square-zero genus bound of Lemma 4.3 for the double. In particular* [LP25, Lemma 7] *holds verbatim for* W', *and the slicing obstruction for* 4_1 *in* W' *persists.*

Part I also makes the obstruction quantitative, and shows that Theorem D is not a routine computation. In Section 3 we build a fully explicit finite presentation of $\pi_1(R)$ with no genus-2 mapping class computations: since Q is the square knot, the closed genus-2 fiber splits along the connect-sum circle and the entire bundle structure is generated by the trefoil monodromy

$$h\colon x \mapsto y^{-1}, \qquad y \mapsto yx$$

on $F_2 = \langle x, y \rangle$, together with the handle swap. With that model, Section 6 shows that all 72 *uncorrected* candidate relation systems for $\pi_1(V')$ collapse to the trivial group, and that this collapse is over-determined, carried entirely by four monodromy relations broken by the surgery tori, so that the value of $\pi_1(V')$ rests on the based-loop correction words alone. A direct sensitivity experiment sharpens this: across 120 sampled candidate correction words, 69 present the trivial group, 44 a group not visibly trivial, and the remaining 7 are detectably wrong ($H_1 \neq 0$). *Both nondegenerate outcomes are generic*, so a trivial outcome obtained from guessed words is worthless. The same diagnosis applies to the presentation printed in [AP10, Lemma 8]: five parameter values give the groups predicted by their Theorem 9, while this finite check does not prove the all-parameter assertion (Section 6.4). It does show in those instances why the geometric derivation of the based words, rather than the subsequent coset enumeration, is the step requiring scrutiny.

Part II (Sections 7–12) computes the group to the standard this diagnosis demands. Our model is the approach of Baldridge and Kirk [BK08, BK09], who write of their own Luttinger-surgery computations that "the introduction of unwanted conjugation at any stage can easily lead to a loss of control over fundamental groups, in particular leading to plausible but unverifiable calculations" [BK09, Section 1]. Concretely, we work in a single explicit model (the fiber is a regular octagon with its edges identified, the base a cut square), chosen so that every monodromy lift is a genuinely based automorphism (the basepoint is fixed by both monodromies, so no basing correction is ever implicit); the based meridians and Lagrangian push offs of the two surgery tori are computed as explicit words in the fiber and base generators (Section 8); an additional clean relation is obtained from an explicit basis of a drilled fiber, and the resulting true-relation system is proved to surject onto the fundamental group (Section 10); and the relation systems are decided by coset enumeration and, where enumeration does not terminate, by Knuth–Bendix completion (Section 11). Theorem D is proved in Section 11, and Theorems A–C are deduced from it and Theorem 1.2 in Section 12.

Three safeguards back the computation, and all three live in the appendices so as not to interrupt the mathematics. First, Lemma 7.1 and Figure 1 fix the geometric curves. Given those geometric inputs, every input word (each meridian, each push off, and each corrected monodromy relation) is independently developed from its departure, crossing, and arrival data by a small algorithm whose validation is described in Appendix A. The algorithm checks the reading of a specified curve; it does not identify the source curve. Second, the fixed-V robustness family contains all 4,096 sign, arc and placement choices described above, and every resulting relation group is trivial (Remark 11.1; Appendix A). The separate 1,408-system grids, including a second based diagram, are diagnostics; their pre-R_3 overflows carry no conclusion. Third, the whole method was calibrated on two configurations with independently known answers: the T^4 double-Luttinger configuration of Baldridge–Kirk [BK09], whose complement presentation they computed with explicit care about basings, and a Seifert-fibered configuration whose surgeries are torus twists with classified fundamental groups (Section 8.6; Appendix B). The T^4 calibration is not decorative: the first version of this derivation contained one wrong input word, the calibration caught it, and the repaired word, with every conclusion re-established, is the one derived in Section 8.5.

Ancillary files. The development outputs, run counts, and finite-group decisions reproduce from the ancillary files accompanying the arXiv submission (also available at `https://github.com/bwuebben/exotic-s2xs2`); total runtime is minutes on a laptop (GAP 4.16 [GAP], Python 3), except one optional finite-quotient search (≈ 56 CPU-hours). An additional `walkthrough.pdf` is available in the GitHub repository's `papers/` directory; it is not part of the arXiv ancillary archive. Appendix A inventories the files and separates the geometric arguments proved in the body from the computations reproduced by the package.

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2. Luttinger surgery, basings, and conventions

2.1. **Luttinger surgery and the fundamental group.** We use Luttinger surgery in the form of [Lut95, ADK03], with the π_1-bookkeeping of [BK09, Section 2], which we recall. For a Lagrangian torus T in a symplectic 4-manifold M, the Darboux–Weinstein theorem provides a parameterization $T^2 \times D^2 \to \nu(T) \subset M$ under which $T^2 \times \{d\}$ is Lagrangian for every $d \in D^2$; choosing $d \neq 0$ gives the *Lagrangian push off* $F_d \colon T \to T^2 \times \{d\} \subset M - T$, and the isotopy class in $\nu(T) - T$ of the push off $F_d(\gamma)$ of an embedded curve $\gamma \subset T$, its *Lagrangian framing*, depends only on the symplectic structure. A curve isotopic to $\{t\} \times \partial D^2$ is a *meridian* of T, denoted μ. The $1/k$ *Luttinger surgery on* T *along* γ removes $\nu(T)$ and reglues it so that the curve $\mu F_d(\gamma)^k$ bounds a disk; the result is again symplectic, and its fundamental group is

$$\pi_1(M - T)/\langle\langle \mu\, F_d(\gamma)^k \rangle\rangle,$$

where $\langle\langle\cdot\rangle\rangle$ denotes the normal closure. Following [BK09, Section 2], when the basepoint of M lies off $\partial\nu(T)$ the based loops μ and $F_d(\gamma)$ are to be joined to the basepoint *by the same path* in $M - T$, and the displayed formula holds with respect to that choice of basing. All the delicacy of this paper lives in that sentence: the surgered group depends on the based words of μ and $F_d(\gamma)$, not on their free homotopy classes, and Section 6 quantifies how badly free-homotopy reasoning fails here. In the constructions of [BH23] this delicacy is engineered away: the meridians of the surgered tori are recorded only as conjugates of commutators, the exact expressions being unnecessary ([BH23, Remark 3]), because each meridian is used only in a quotient where its commutator core already dies, so the conjugating words never enter. No such collapse is available for V: the presentations of Part II must be decided whole, and every basing arc and conjugating path is recorded for that reason.

Both of our surgeries have $k = \pm 1$; which sign corresponds to which geometric convention is one of the five signs of Convention 2.1(C7), and no conclusion depends on it.

2.2. **Conventions, collected.** Every convention used in this paper is collected here; each is restated in context where it is first used. Two of them point in opposite directions and are the easiest place to lose one's footing, so they are put side by side.

Convention 2.1 (Collected)**.** (C1) *Paths and loops compose left to right*: $\gamma_1\gamma_2$ traverses γ_1 first. Relators are words equal to 1; a relation written $ugu^{-1} = w$ means the relator $ugu^{-1}w^{-1}$.

- (C2) *Automorphisms act on the left and compose right to left*: fg means "g first". In particular $h := T_a \circ T_b$ applies T_b first. *(C1) and (C2) are deliberately opposite*: the first is a statement about traversal, the second about function composition.
- (C3) *Base generators*: $A := [\bar\alpha]^{-1}$ and $B := [\bar\beta]^{-1}$, so that conjugation by B implements one *upward* transport around $\bar\beta$, and likewise A for $\bar\alpha$ (Convention 7.2).
- (C4) *Cut square*: base $=$ $[0,1]^2$ minus a puncture disk at $(3/4, 3/4)$, basepoint $q =$ $(1/4, 1/4)$; crossing the α-cut *upward* applies ψ_0, crossing the β-cut *rightward* applies φ_0 (Section 7.3). The surgery tori sit on the cuts: $T_\alpha = c \times \{\alpha\text{-cut}\}$, $T_\beta = e \times \{\beta\text{-cut}\}$.
- (C5) *Twist normalization*: $T_a \colon (x, y) \mapsto (x, yx)$, $T_b \colon (x, y) \mapsto (xy^{-1}, y)$; the residual chirality ambiguity is absorbed by the handle swap (Remark 3.3).
- (C6) *Basing arcs and conjugating paths*: every meridian and push off is joined to the basepoint by a stated basing arc, per Section 2.1; when a monodromy relation acquires a meridian correction, the correction enters conjugated by an explicit path (the conjugating path of Section 8.3).

(C7) *The five signs.* ε_3 is the meridian sign of the corrected As relation, ε_4 that of By, and ε that of Bs (the bare letter is reserved throughout for this one sign, which reappears anti-coupled in Section 8.5); $\varepsilon_A, \varepsilon_B \in \{\pm 1\}$ are the exponents of the two surgery relations. These are the $\pm$ signs written unnamed in Sections 8.3 and 9; every conclusion below holds for all 32 assignments (Remark 11.1).

(C8) *Dictionary with* [LP25]: $x \sim a$, $y \sim b$, $r \sim e$, $s \sim d$, and $c \sim xr$. With compatible orientations, the auxiliary source curve has $[z] = [y] - [s]$; it is not used in the Part II relation system (Sections 3.2 and 8.1).

One fixed convention set is used throughout the body of the paper. The finite check includes every sign and arc choice that can alter the displayed relator sheet, as well as deliberate left/right perturbations. Reversing the composition convention or the chirality assignment instead gives the generator changes described in Remarks 3.7 and 3.3.

Part I. The reduction

3. The explicit model of $\pi_1(R)$

We recall the construction of [LP25, Section 1]. The 0-surgery $S^3_0(Q)$ on the square knot $Q = 3_1\#\overline{3_1}$ fibers over S^1 with closed genus-2 fiber F and monodromy conjugate to $abd^{-1}e^{-1}$, a positive Dehn twist along each of the chain curves a, b, c, d, e of [LP25, Figure 1] being denoted by the same letter. Let φ be the involution of F exchanging $a \leftrightarrow e$, $b \leftrightarrow d$ and preserving c setwise. Since $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$ in the mapping class group, the F-bundle R over the once-punctured torus with monodromy ab along β and φ along α has $\partial R = S^3_0(Q)$.

Convention 3.1. $[u, v] = uvu^{-1}v^{-1}$. Automorphisms act on the left and compose right-to-left: fg means "g first". Mapping classes act on π_1 only up to inner automorphism; every presentation below depends on a choice of automorphism lifts, see Remark 3.6.

3.1. The trefoil monodromy.

Let $\Sigma_{1,1}$ be a once-punctured torus with basepoint on the boundary, $\pi_1(\Sigma_{1,1}) = F_2 = \langle x, y\rangle$, boundary word $[x, y]$, and let the core curves realizing x, y be $a \simeq x$ and $b \simeq y$, so $a \cdot b = 1$. For a suitable orientation convention the twists act by

$$T_a\colon (x, y) \mapsto (x,\, yx), \qquad T_b\colon (x, y) \mapsto (xy^{-1},\, y).$$

Define $h := T_a \circ T_b$ (T_b first).

Lemma 3.2. *$h(x) = y^{-1}$, $h(y) = yx$, and:*

1. $h([x, y]) = [x, y]$ exactly *(not merely up to conjugacy);*
2. $h^6 = \mathrm{conj}_{[x,y]^{-1}}$*, the full boundary Dehn twist;*
3. h^3 *acts on* H_1 *as* $-\mathrm{id}$*;*
4. *on* $H_1(\Sigma_{1,1}) = \mathbb{Z}^2$*,* h *acts by* $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 1\end{smallmatrix}\right)$*: determinant* 1*, trace* 1*, order* 6*.*

Consequently h is the fibered monodromy of a trefoil: it is a product of single positive twists along dual non-separating curves, fixes the boundary pointwise (reflected in (1)), and is freely periodic of period 6 with h^6 the boundary twist (2), with the hyperelliptic involution at h^3 (3).

Proof. Direct computations in F_2, easily reproduced by hand (and checked by machine; Appendix A). For (1):

$$h([x, y]) = [y^{-1}, yx] = y^{-1} \cdot yx \cdot y \cdot x^{-1} y^{-1} = xyx^{-1}y^{-1} = [x, y].$$

□

Remark 3.3 (Chirality)**.** Whether h or h^{-1} ($h^{-1}\colon x \mapsto xy,\ y \mapsto x^{-1}$) is the right-handed trefoil is a convention we will not need: Q is the connected sum of a trefoil and its mirror, and exchanging the chirality assignment amounts to the handle relabeling $(x, y) \leftrightarrow (r, s)$ below, under which every construction and every computation in Section 6 is symmetric.

3.2. **The closed fiber and the bundle group.** The fiber of $Q = 3_1 \# \overline{3_1}$ is the boundary connected sum of the two trefoil fibers, and the closed fiber F of $S^3_0(Q)$ splits along the connect-sum circle into two once-punctured tori. Writing $\pi_1(F) = \langle x, y, r, s \mid [x, y][r, s]\rangle$ with (x, y) carried by the left handle and (r, s) by the right, the dictionary with [LP25, Figure 1] is

$$a \sim x, \quad b \sim y, \quad e \sim r, \quad d \sim s, \qquad c \sim xr, \qquad [z] = [y] - [s],$$

where $[c] = [a] + [e]$. The sign in the last formula is forced by the source figure: z is disjoint from c and meets a and e once, so the two handle contributions have opposite signs. The diagnostic calculations of Section 6 use ys^{-1} and $s^{-1}y$ as representative candidate words with this homology class; no based word for z enters the proof of Theorem D. The matching $\sim$ is orientation-insensitive: it records which embedded curve realizes which free-homotopy class; the oriented based developments of Part II invert a and e (validation V3, Appendix A). Define

$$\tilde{\varphi}:\ x \leftrightarrow r,\ y \leftrightarrow s \qquad \text{and} \qquad \tilde{\psi} := h * \mathrm{id}\colon\ (x, y, r, s) \mapsto (y^{-1},\ yx,\ r,\ s),$$

lifting φ and ab respectively; $\tilde{\psi}$ is well defined on the closed-fiber group because h fixes the boundary word exactly (Lemma 3.2(1)).

Lemma 3.4. (1) *$\tilde{\psi}$ fixes the relator $[x, y][r, s]$ exactly; $\tilde{\varphi}$ sends it to its $[x, y]$-conjugate. In particular both define automorphisms of $\pi_1(F)$.*
(2) *$\tilde{\varphi}^2 = \mathrm{id}$, and $\tilde{\psi}\,\tilde{\varphi}\,\tilde{\psi}^{-1}\,\tilde{\varphi}^{-1} = h * h^{-1}\colon (x, y, r, s) \mapsto (y^{-1},\ yx,\ rs,\ r^{-1})$.*

Proof. Direct computation in the free group (checked by machine; Appendix A); (2) also follows from block algebra: $\tilde{\varphi}(h * \mathrm{id})^{-1}\tilde{\varphi} = \mathrm{id} * h^{-1}$, so the commutator is $(h * \mathrm{id})(\mathrm{id} * h^{-1}) = h * h^{-1}$. □

Lemma 3.4(2) realizes the mapping-class factorization $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$ of [LP25] as a literal identity of automorphisms: $h * h^{-1}$ is the 0-surgery monodromy of the square knot in this model (the left handle twisted by the trefoil, the right by its mirror).

Proposition 3.5 (The presentation)**.**

$$\pi_1(R) = \Big\langle x, y, r, s, \alpha, \beta \,\Big|\, [x, y][r, s],\ \alpha g \alpha^{-1} = \tilde{\varphi}(g),\ \beta g \beta^{-1} = \tilde{\psi}(g) \quad (g \in \{x, y, r, s\})\Big\rangle.$$

Proof. R fibers over the once-punctured torus T_0 with aspherical fiber F; the long exact sequence of the fibration reduces to $1 \to \pi_1(F) \to \pi_1(R) \to \pi_1(T_0) \to 1$, and $\pi_1(T_0) = F_2\langle \alpha, \beta\rangle$ is free, so the sequence splits and $\pi_1(R) = \pi_1(F) \rtimes F_2$ with respect to the chosen automorphism lifts $\tilde{\varphi}, \tilde{\psi}$ of the two monodromies. The displayed presentation is the standard presentation of such a semidirect product. □

Remark 3.6 (Lift freedom; where the based-loop indeterminacy lives)**.** A mapping class determines its action on $\pi_1(F)$ only up to inner automorphisms. Replacing $\tilde{\varphi}, \tilde{\psi}$ by other lifts changes the presentation of Proposition 3.5 by the substitution $\alpha \mapsto \alpha w$, $\beta \mapsto \beta w'$ ($w, w' \in \pi_1(F)$), an isomorphism of $\pi_1(R)$, but one that changes the *words* that geometric objects (meridians, surgery directions) are represented by. All presentations of the surgered manifolds in Section 6 are therefore *candidates*, well defined only once based representatives are fixed; the dependence of $\pi_1(V')$ on that choice is precisely the point of the second experiment (Section 6.3). Fixing those representatives is the business of Part II.

Remark 3.7 (Convention robustness)**.** Replacing the monodromy lift $\tilde{\psi}$ by $\tilde{\psi}^{-1}$ (the holonomy-direction convention) is the substitution $\beta \mapsto \beta^{-1}$, which converts the surgery relation $[z, \alpha]^{\varepsilon}\beta = 1$ into $[z, \alpha]^{-\varepsilon}\beta = 1$: an ε-flip, and the computational grid of Section 6 covers all ε. Together with Remark 3.3, the collapse of the LP cases below is independent of every orientation and composition convention made here; the other diagnostic systems (whose convention choices can also change word order and conjugation) are run in the conventions stated.

4. Rigidity of the surgery parameters

Recall from [LP25, Section 1] the two Lagrangian tori: T_β, the sub-bundle with fiber e over β, and T_α, the sub-bundle with fiber c over α (the monodromy ab fixes e pointwise; φ preserves c with orientation). Their meridians are commutators in the complement $C := R \setminus \nu(T_\alpha \cup T_\beta)$: $\mu_{T_\beta} \simeq [z, \alpha'']$ and $\mu_{T_\alpha} \simeq [d, \beta'']$ (see Remark 4.1), where α'', β'' project to α, β.

Remark 4.1 (The geometric dual of T_α)**.** The proof of [LP25, Lemma 5] prints the dual as (b, β''); this cannot close up into a torus, since ab does not preserve b even homologically (the only twist in ab affecting b is T_a, and $a \cdot b = 1$, so $[ab(b)]$ has $[a]$-coefficient ± 1 in every convention). The curve dual to c that ab *does* fix (pointwise, being disjoint from $a \cup b$, exactly as [LP25, p. 3] observe for e) is d; the dual is therefore $T_\alpha^* = (d, \beta'')$, with μ_{T_α} freely homotopic to $[d, \beta'']$. Since φ exchanges $b \leftrightarrow d$, the confusion is a natural one, and no result of [LP25] is affected: their arguments use only that each meridian is a commutator of a fiber curve with a curve projecting to the base. At the π_1 level, however, the correction matters, and it is used throughout this paper.

Lemma 4.2. *Let $C = R \setminus \nu(T_\alpha \cup T_\beta)$. Then the inclusion induces an isomorphism*

$$H_1(C) \xrightarrow{\cong} H_1(R) = \mathbb{Z}^2\langle \alpha', \beta' \rangle.$$

In particular every fiber class vanishes in $H_1(C)$.

Proof. First, $H_1(R) \cong \mathbb{Z}^2$: by Proposition 3.5, $H_1(R)$ is the direct sum of $\mathbb{Z}^2\langle \alpha, \beta \rangle$ and the coinvariants of $H_1(F) = \mathbb{Z}^4$ under the subgroup generated by $\tilde{\varphi}_*$ and $\tilde{\psi}_*$; writing Φ, Ψ for the matrices of $\tilde{\varphi}_*, \tilde{\psi}_*$, the stacked matrix $(\Phi - I; \Psi - I)$ has Smith normal form $\mathrm{diag}(1, 1, 1, 1)$ (a direct computation, also checked by machine), so the coinvariants vanish. (Concretely: $\Psi - I$ is invertible on the (x, y)-block, killing x, y; then the swap relations kill r, s.)

For the complement, excision and the Künneth formula for $(\nu(T), \partial\nu(T)) \cong T^2 \times (D^2, \partial D^2)$ give $H_1(R, C) = 0$ and $H_2(R, C) \cong \mathbb{Z}^2$, generated by the two normal disks, whose boundaries are the meridians. The homology sequence of the pair,

$$H_2(R, C) \xrightarrow{\partial} H_1(C) \to H_1(R) \to H_1(R, C) = 0,$$

therefore presents $H_1(R)$ as the quotient of $H_1(C)$ by the images of the meridians; each meridian is a commutator in $\pi_1(C)$, hence trivial in $H_1(C)$, so the map $H_1(C) \to H_1(R)$ is an isomorphism. $\square$

Proof of Proposition 1.3. Forcing. A $1/k$ Luttinger surgery on T_β along an embedded direction, i.e. a primitive class $p\beta' + qe$ on T_β ($\gcd(p,q) = 1$), fills the boundary of $\nu(T_\beta)$ so that the class $\mu_{T_\beta} + k(p\beta' + qe)$ dies (orientation conventions are absorbed into the sign of k). By Lemma 4.2, in H_1 this relation reads $kp\,\beta' = 0$, since the meridian is a commutator and e is a fiber class. Together with the corresponding relation $k'p'\,\alpha' = 0$ from T_α,

$$H_1(V') \;=\; \mathbb{Z}^2/\langle kp\,\beta',\, k'p'\,\alpha'\rangle \;=\; \mathbb{Z}/|kp| \oplus \mathbb{Z}/|k'p'|,$$

which vanishes iff $|kp| = |k'p'| = 1$: the coefficients are ±1 and the directions have base component ±1, i.e. equal β', α' modulo fiber classes. Purely-fiber directions ($p = 0$) impose no relation and leave $H_1 = \mathbb{Z}^2$.

Realization. Conversely, for coefficients ±1 and directions $\beta' e^m, \alpha' c^n$ the same computation gives $H_1(V') = 0$.

Preserved structure. Each direction above is a primitive class realized by an embedded curve on the corresponding Lagrangian torus, so the surgeries are Luttinger surgeries and the result carries a symplectic structure [Lut95, ADK03]; χ is unchanged. The argument of [LP25, Lemma 5] now applies verbatim: $H_1(V') = 0$ and $\partial V'$ connected give $H_3(V') = 0$ and $H_2(V')$ free; $\chi(V') = 2$ gives $H_2(V') \cong \mathbb{Z}$, generated by a fiber F disjoint from all surgery tori; $F^2 = 0$; since $H_1(V') = 0$ we get $H_2(V'; \mathbb{Z}/2) \cong \mathbb{Z}/2\langle F\rangle$, and evaluation against it detects $H^2(V'; \mathbb{Z}/2)$, so $\langle w_2, F\rangle = F \cdot F = 0$ forces $w_2(V') = 0$: V' is spin. All surgeries take place in $\operatorname{int} V$, so $\partial V' = \partial V = S^3_0(Q)$ and the free involution σ descends; the proof of [LP25, Lemma 7] applies verbatim and $W' = V'/\sigma$ is a spin rational homology 4-sphere with $H_1(W') = \mathbb{Z}/2$.

Persistence of the obstruction input. The double $Z' = V' \cup_\sigma V'$ is obtained from the genus-2 bundle $R \cup_\sigma R$ over a genus-2 surface by Luttinger surgeries on disjoint Lagrangian tori; the fiber F and the section $\hat{\Gamma}$ coming from a fixed point of φ ([LP25, Lemma 6]) form a hyperbolic pair, and all surgery tori miss a fiber (they lie over curves in the base, which miss a point) and miss the two fixed-point sections (the fiber curves c, e can be isotoped off the two fixed points of φ, exactly as in [LP25, Lemma 6]). The Thurston form can be chosen so that both F and $\hat{\Gamma}$ are symplectic: its fiber term is positive on F, and a sufficiently large base term is positive on the section. Luttinger surgery leaves the form unchanged away from the surgery neighborhoods, so these two surfaces remain symplectic for every admissible coefficient and direction. Lemma 4.3 turns precisely these facts into the slicing obstruction; no generality beyond its stated hypotheses is attributed to [SS23]. $\square$

Lemma 4.3 (No square-zero tori)**.** *Let V' be an admissible variant and suppose that its double $Z' = V' \cup_\sigma V'$ is simply connected. No nonzero square-zero class in $H_2(Z'; \mathbb{Z})$ is represented by a smoothly embedded torus.*

Proof. The double is symplectic. Since $\chi(Z') = 4$ and $\pi_1(Z') = 1$, its second homology has rank 2; the classes $F, \hat{\Gamma}$ span it because their intersection matrix below is unimodular. Thus $H_2(Z'; \mathbb{Z}) = \mathbb{Z}\langle F, \hat{\Gamma}\rangle$ with

$$F^2 = \hat{\Gamma}^2 = 0, \qquad F \cdot \hat{\Gamma} = 1,$$

where both F and $\hat{\Gamma}$ are symplectic surfaces of genus 2 by the preceding paragraph. Here $F^2 = 0$ is the bundle framing. For the section, use the explicit gluing of [LP25, Lemmas

4 and 6] and choose the section through p. In the local model of Section 7.1, the two monodromies have derivatives $D\psi_0|_p = I$ and $D\varphi_0|_p = -I$: the twists are supported away from p, while φ_0 is the half-turn. Thus the vertical normal bundle on either punctured-torus half carries a flat $SO(2)$ structure with these two holonomies. Its boundary holonomy is their commutator, hence is exactly I, and parallel transport supplies a boundary framing. This framing extends over the half: capping the boundary by a trivial flat disk gives the flat $SO(2)$ bundle on a torus defined by the commuting rotations $I, -I$, whose Euler class is zero. Use these extending framings on both halves. The fiber component of σ is the single diffeomorphism appearing in [LP25, Lemma 4], independent of the boundary parameter, and its derivative at p is therefore a constant linear clutching map in these framings. The associated map $S^1 \to GL^+(2, \mathbb{R})$ has degree zero, so the closed normal bundle has Euler number zero and $\hat{\Gamma}^2 = 0$. The surfaces meet once because one is a fiber and the other a section.

For each $k > 0$, choose a smooth map $f : F \to S^1$ inducing an epimorphism on H_1 and, in $F \times D^2$, take

$$\Sigma_k = \{(u, z) : z^k = \epsilon f(u)\},$$

where $S^1 \subset \mathbb{C}$ and $\epsilon > 0$ is small. This is an embedded, unbranched degree-k cover of F. It is connected because the monodromy of the cover is the reduction of $f_* : \pi_1(F) \twoheadrightarrow \mathbb{Z}$ modulo k, hence is transitive. It therefore has genus $k + 1$ and represents kF. For sufficiently small ϵ, its projection to F is locally orientation-preserving and the fiber-area term is positive on it, so Σ_k is symplectic. The same construction applies to $\hat{\Gamma}$. By the symplectic Thom theorem [OS00], these surfaces minimize genus in the classes kF and $k\hat{\Gamma}$. Reversing the orientation of a representative does not change its genus, hence

$$g_{Z'}(kF) = g_{Z'}(k\hat{\Gamma}) = |k| + 1 \qquad (k \neq 0).$$

This is the argument used for the square-zero axes in [SS23, proof of Theorem 1.4], written here with its actual hypotheses.

If $u = aF + b\hat{\Gamma}$, then $u^2 = 2ab$. Thus a nonzero square-zero class is kF or $k\hat{\Gamma}$ for some $k \neq 0$, and its minimal genus is $|k| + 1 \geq 2$. It therefore has no torus representative. □

5. The reduction theorem

Proof of Theorem 1.2. (a). The double cover $Z' \to W'$ gives a short exact sequence $1 \to \pi_1(Z') \to \pi_1(W') \to \mathbb{Z}/2 \to 1$, with $\pi_1(Z') \hookrightarrow \pi_1(W')$ *injective* by covering theory.

($\Leftarrow$) If $\pi_1(Z') = 1$ then $\pi_1(W') \cong \mathbb{Z}/2$. Now W' and B are closed, oriented, *smooth* spin 4-manifolds with $\pi_1 = \mathbb{Z}/2$ (Proposition 1.3 for W'; [Kaw09, LP25] for B) and both are rational homology spheres, so both intersection forms on H_2/tors are trivial, both w_2-types are type (II), and both Kirby–Siebenmann invariants vanish (smoothness; alternatively $\text{KS} \equiv \sigma/8 \pmod 2$ for spin topological 4-manifolds). By the classification of closed oriented topological 4-manifolds with finite cyclic fundamental group ([HK93, Theorem C]: π_1, the intersection form on H_2/Tors, the w_2-type and KS are a complete set of invariants, extending [HK88, Theorem B] from odd order; see also [Ham08, Theorem 5.1] and [Tei92]) we get $W' \cong_{\text{homeo}} B$. In fact, by the $b_2 = 0$ case as stated in [BSS24] there are exactly two such topological manifolds for each cyclic group, one spin and one non-spin; the spin one is the spun lens space, so $W' \cong_{\text{homeo}} B \cong_{\text{homeo}} L_2$ in the notation of [BSS24].

($\Rightarrow$) A homeomorphism gives $\pi_1(W') \cong \pi_1(B) = \mathbb{Z}/2$, and the injection above forces $|\pi_1(Z')| = 1$.

(b). Assume $\pi_1(Z') = 1$.

Step 1: Z' is homeomorphic to $S^2 \times S^2$. Z' is closed, simply connected, spin (it double covers the spin manifold W'; alternatively glue the spin structures of the two copies of V'), with $\chi(Z') = 2\chi(V') - \chi(S^3_0(Q)) = 4$, so $b_2(Z') = 2$. The classes of the fiber F and of the closed section $\hat{\Gamma} = \Gamma \cup_\sigma \Gamma$ satisfy $F^2 = 0$ and $F \cdot \hat{\Gamma} = 1$; since the form is even (spin), a change of basis $\hat{\Gamma} \mapsto \hat{\Gamma} - \frac{1}{2}(\hat{\Gamma}^2)F$ exhibits the intersection form as the hyperbolic form H. By Freedman's classification [Fre82], the form H, together with the vanishing Kirby–Siebenmann invariant of the smooth manifold Z', determines the homeomorphism type; hence $Z' \cong_{\text{homeo}} S^2 \times S^2$.

Step 2: Z' is not diffeomorphic to $S^2 \times S^2$. This is the argument of [LP25, Theorem 8], which is π_1-agnostic: $R \cup_\sigma R$ is a non-trivial genus-2 surface bundle over a genus-2 surface, hence aspherical, hence minimal and neither rational nor ruled, so its Kodaira dimension satisfies $\kappa \neq -\infty$; Luttinger surgery preserves κ [HL12], and Z' is minimal (it is spin), so $\kappa(Z') \neq -\infty$. Since κ is a diffeomorphism invariant [Li06] and $\kappa(S^2 \times S^2) = -\infty$, Z' is not diffeomorphic to $S^2 \times S^2$.

Step 3: the pair (B, W'). By (a), $W' \cong_{\text{homeo}} B$. The knot 4_1 is slice in B by construction [Kaw09, LP25]. It is not slice in W': following [LP25, Section 2], since W' is spin the relative Rokhlin theorem [Klu21, Theorem 2] and $\mathrm{Arf}(4_1) = 1$ rule out a null-homologous slice disk; a homologically essential slice disk makes the 0-trace $X_0(4_1)$ embed in W' with H_2-essential image, producing a square-zero essential torus $T \subset W'$ (cap a Seifert surface with the disk); since $X_0(4_1)$ is simply connected, T lifts to an essential square-zero torus in Z'; and Proposition 1.3 verifies for Z' the hypotheses of Lemma 4.3, which forbids such a torus. Sliceness of 4_1 distinguishes the smooth structures, so W' is homeomorphic and not diffeomorphic to B.

(c). Assume $\pi_1(V') = 1$. The double Z' and the reglued manifold $Z'' = V' \cup_{f \circ \sigma} V'$ (with f the boundary self-diffeomorphism of [LP25, Figure 3]) are unions of two copies of V' along $S^3_0(Q)$; by van Kampen each fundamental group is a quotient of $\pi_1(V') * \pi_1(V') = 1$, so $\pi_1(Z') = \pi_1(Z'') = 1$ and parts (a)–(b) apply. (This is where the hypothesis $\pi_1(V') = 1$ is genuinely used: the regluing twists the van Kampen data by f_*, and no implication from $\pi_1(Z') = 1$ alone is claimed; see Remark 5.1.) The framing-parity change makes Z'' non-spin with $b_2 = 2$ and signature 0, so its intersection form is $\langle 1 \rangle \oplus \langle -1 \rangle$ and Freedman gives $Z'' \cong_{\text{homeo}} \mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. The relative invariants $\Psi_{V', \mathfrak{t}_\pm}$ are non-zero exactly as in [LP25, Lemma 9]: Z' is symplectic and its canonical class evaluates ± 2 on a fiber disjoint from the surgery tori (Luttinger surgery does not change this pairing [ADK03, HL12]), so [OS04] applies as there (cf. [Thu76]). By [LP25, Lemma 10] the mixed invariant of Z'' for the line $\mathrm{span}\{F\}$ is non-zero; since $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$ (indeed $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2\#D$ for any homology sphere D) admits, for either square-zero line, a square-zero splitting along $S^2 \times S^1$ forcing all mixed invariants to vanish ([LP25, proof of Theorem 2], using $HF_{red}(S^2 \times S^1) = 0$), Z'' is not diffeomorphic to $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. □

Remark 5.1. The implication $\pi_1(V') = 1 \Rightarrow \pi_1(Z') = 1$ is immediate from van Kampen; we do not know whether the converse holds in this setting, since in the amalgam $\pi_1(Z') = \pi_1(V') *_{\pi_1 S^3_0(Q)} \pi_1(V')$ the edge maps need not be injective. The distinction affects only part (c): parts (a) and (b) use $\pi_1(Z')$ alone, while the regluing twists the van Kampen data by f_* and is controlled here through $\pi_1(V')$. It is in any case immaterial in practice: any proof of simple connectivity in this frame proceeds by computing $\pi_1(V')$, which is what Part II does.

Corollary 5.2. *The homeomorphic upgrade of* [LP25, Theorem 1] *within the Lidman–Piccirillo setting is equivalent to the exotic $S^2 \times S^2$ problem: it holds for some admissible* *variant if and only if some admissible double Z' is an exotic $S^2 \times S^2$. Conversely, exhibiting $\pi_1(V') = 1$ for any admissible variant yields simultaneously an exotic $S^2 \times S^2$, a homeomorphic-but-not-diffeomorphic pair distinguished by unconstrained slicing, and a simply connected exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.*

6. The obstruction is based-loop data

All computations in this section are coset enumerations and abelianizations in GAP 4.16 [GAP]; the scripts are ancillary files (Appendix A), and every run below reproduces in seconds on a laptop. The enumeration cap is 4×10^5 cosets; a run that exceeds it is recorded as *not visibly trivial*, never as "nontrivial".

6.1. **Candidate presentations.** By Proposition 3.5 and Remark 3.6, a presentation of $\pi_1(V')$ requires, beyond the exact bundle relations, based representatives for the two surgery relations. The *uncorrected candidates* take the naive words suggested by the free homotopy types: filling relations

$$[z, \alpha]^{\varepsilon_1} \cdot \beta\, r^m = 1, \qquad [w, \beta]^{\varepsilon_2} \cdot \alpha\, (xr)^n = 1,$$

with $z \in \{ys^{-1}, s^{-1}y\}$, $\varepsilon_i \in \{\pm 1\}$, mixed-direction exponents m, n as in Proposition 1.3, and, in the parallel-tori scheme, the additional fiber-direction relations $[z, \alpha]^{\varepsilon_3} r = 1$, $[w, \beta]^{\varepsilon_4}(xr) = 1$. The grid of Section 6.2 takes $w = y$, the dual printed in [LP25]; the corrected dual $w = s$ of Remark 4.1 is run separately.

6.2. **The uncorrected candidates collapse universally.** Over all 8 conventions for the LP surgeries, the 32 mixed-direction cases ($m, n \in \{-1, 1\}$), and the 32 parallel-tori cases, *for every one of the* 72 *uncorrected candidate groups the enumeration completes with index* 1*: the trivial group* (Appendix A; under a second), while the controls do not (the bundle group alone has $H_1 = \mathbb{Z}^2$, and either surgery alone leaves $H_1 = \mathbb{Z}$). This universal collapse does not validate the candidates: they retain every unsurgered monodromy relation without first checking whether its transport annulus survives in the torus complement. If all those relations were valid, then $\pi_1(V) = 1$ and Theorem 1.2 would give an exotic $S^2 \times S^2$; the fact that [LP25] does not make that claim is historical context, not a mathematical contradiction. The actual defect is geometric and can be localized directly. A monodromy relation $\beta g \beta^{-1} = \tilde{\psi}(g)$ is carried by a transport annulus $g \times \beta$; in the surgered manifold the relation holds only if the annulus avoids the surgery tori. Intersection numbers in R decide this: an annulus over one base curve meets a sub-bundle torus over the transverse base curve above their intersection point, and meets a torus over the *same* base curve along it; in both cases the count is the intersection number of the fiber curves. Using $[c] = [a] + [e]$:

relation carried over α or β	obstruction
$x\ (\sim a),\ r\ (\sim e)$ over α, β	$a \cdot c = a \cdot e = e \cdot c = 0:$ clean
$y\ (\sim b)$ over β	$b \cdot c = 1:$ crosses T_α
y over α	$b \cdot c = 1:$ crosses T_α
$s\ (\sim d)$ over α	$d \cdot c = 1:$ crosses T_α
s over β	$d \cdot e = 1,\ d \cdot c = 1:$ crosses T_β, T_α

Exactly the four relations for $y \sim b$ and $s \sim d$ are broken; in V' they hold only with insertions of conjugates of words carried by the glued-in tori (c-, e-, α'-, β'-words). The four relations for x and r are clean.

Remark 6.1 (Embedded count versus based count)**.** The count above is that of transport annuli over the *embedded* curves. Like the correction words themselves, it is diagram data. In the based diagram of Part II the base loops meet the cut circles minimally ($\bar{\alpha}$ never meets the α-cut and $\bar{\beta}$ never meets the β-cut), and two of the four crossings predicted above disappear: the transport annulus of y over $\bar{\alpha}$ meets no surgery torus at all ($\bar{\alpha}$ misses the α-cut over which T_α sits, and y misses e), so that relation is clean there; and the annulus of s over $\bar{\beta}$ misses T_β, keeping only its single T_α-crossing. Exactly *three* relations acquire corrections in that diagram, each at a single crossing (Remark 7.3).

6.3. **The corrections decide the group.** Dropping the four broken relations outright leaves a group with $H_1 = \mathbb{Z}^2$ (Appendix A): the broken relations are indispensable already for $H_1(V') = 0$, so no surjection from a clean-relations group onto the actual $\pi_1(V')$ can certify anything: the clean group is not even a homology-correct approximation.

A direct sensitivity experiment sharpens the point. Sampling candidate correction words in the format above (conjugates of the meridians and of the words carried by the glued-in tori, inserted at the broken relations), the same computation presents 120 candidate groups: for 69 the enumeration completes with index 1, for 44 it exceeds the cap, and the remaining 7 are detectably wrong ($H_1 \neq 0$); no case produced a nontrivial finite group. *Both nondegenerate outcomes are generic.*

The value of $\pi_1(V')$ — and with it, by Theorem 1.2, the existence of an exotic $S^2 \times S^2$ — is therefore carried entirely by the based-loop correction words, with no robustness in either direction: a trivial outcome obtained from guessed words certifies nothing, a naive enumeration that exceeds the coset cap proves nothing, and nothing short of a genuine based diagram decides the group.

6.4. **Finite checks of the Akhmedov–Park presentation.** The manifold of [AP10] is built from the same involution: their model surface $(\Sigma_2 \times \Sigma_3)/\mathbb{Z}_2$ fibers over Σ_2 with monodromy the same two-fixed-point involution class as φ. Their Theorem 9 asserts $\pi_1(M_n^p) = \mathbb{Z}/p$ from the presentation printed in their Lemma 8. Coset enumeration from that presentation (Appendix A) confirms

$$\pi_1(M_n^p) = \mathbb{Z}/p \quad \text{for} \quad (n,p) \in \{(1,1),(2,1),(3,1),(1,2),(1,3)\}.$$

These five computations agree with [AP10, Theorem 9] at the tested parameters. They are not a proof for arbitrary n, p, and no such conclusion is used in this paper. Their role here is diagnostic: even when a printed presentation produces the predicted group in several nontrivial cases, the geometric derivation of its meridian and Lagrangian-framing words remains a separate question. In [AP10] those words are the six triples of Lemma 7, with the fifth and sixth involving a path that is not a sub-bundle coordinate.

6.5. **What Part II must supply.** The model of Section 3 reduces the obstruction to a finite, fully specified computation: derive, from an explicit based picture of F carrying the five-chain $a, \ldots, e$ and the basepoint, the actual correction words for the broken monodromy relations and the based surgery data, in the style of [AP10, Lemma 7] or Baldridge–Kirk [BK08]. Because the crossing pattern of Section 6.2 is fixed by the intersection combinatorics, a single based diagram parameterizes the entire exponent family $V'_{m,n}$ of Proposition 1.3 at once. Part II carries this out.

Part II. The computation

7. The based model

7.1. **The symmetric octagon.** The fiber F is the regular octagon with edge word $x\,y\,x^{-1}y^{-1}\,r\,s\,r^{-1}s^{-1}$ (edges $E_1,\dots,E_8$, vertices $V_1,\dots,V_8$, all identified to the single point p), so $\pi_1(F,p)=\langle x,y,r,s \mid [x,y][r,s]\rangle$ with the edge loops as generators (Figure 1). Since all eight vertices are identified to p, a based loop transverse to the edges is specified by its departure corner and its ordered edge crossings, and its based word in x,y,r,s is read off by developing in the octagon-tiled universal cover; we call that word the loop's *development* (the computation is mechanized and validated in Appendix A). Let ρ be the rotation by π. It maps $E_i\to E_{i+4}$, respects the identifications, and induces an involution φ_0 of F with two properties used constantly below: φ_0 acts on $\pi_1(F,p)$ by the *exact* swap $x\leftrightarrow r$, $y\leftrightarrow s$ (as a based automorphism, with no conjugation correction from a basing arc), and $\mathrm{Fix}(\varphi_0)=\{p,O\}$ (O the center), matching the two fixed points of the Lidman–Piccirillo involution φ and giving their two sections Γ,Γ' through p and O.

Since φ_0 fixes p and the twist diffeomorphism ψ_0 below is supported away from p, *every monodromy lift in this paper is a genuinely based automorphism*: the presentation of π_1 of the bundle requires no basing correction at all. This eliminates one entire class of based-loop ambiguity before the computation starts.

The next lemma is the bridge from the surface drawn by Lidman and Piccirillo to this polygon. It is stated before any word is read from the octagon so that the later computation does not silently replace their marked surface by a different one.

Lemma 7.1 (Equivariant normal form)**.** *Let $(F_{\mathrm{LP}};a_{\mathrm{LP}},b_{\mathrm{LP}},c_{\mathrm{LP}},d_{\mathrm{LP}},e_{\mathrm{LP}},\varphi)$ be the marked genus-2 surface of* [LP25, Figure 1]. *There is an orientation-preserving diffeomorphism*

$$q: F_{\mathrm{LP}}\longrightarrow F$$

which conjugates φ to the half-turn ρ and carries the five-chain, in order, to the solid curves drawn in Figure 1. The two fixed points of φ go to p and O. Consequently the polygonal curves used below are ambient representatives of the Lidman–Piccirillo curves, not merely curves with the same homology classes.

Proof. The ordered curves $(a_{\mathrm{LP}},b_{\mathrm{LP}},c_{\mathrm{LP}},d_{\mathrm{LP}},e_{\mathrm{LP}})$ form a five-chain: consecutive curves meet once and all other pairs are disjoint. A regular neighborhood N of their union has genus 2. Indeed, the two disjoint pairs $(a_{\mathrm{LP}},b_{\mathrm{LP}})$ and $(d_{\mathrm{LP}},e_{\mathrm{LP}})$ give two symplectic pairs in $H_1(F_{\mathrm{LP}})$, so N already has genus at least 2. The chain graph has four vertices and eight edges, hence $\chi(N)=-4$; it follows that N has two boundary components. Since $\chi(F_{\mathrm{LP}}\smallsetminus \mathrm{int}\,N)=2$ and those are its only two boundary circles, the complement consists of two disks. Thus the five-chain is a filling ribbon graph.

Orient the five curves successively so that the four intersection signs agree with those of the polygonal chain. The resulting oriented ribbon graphs are isomorphic. The isomorphism thickens to an orientation-preserving diffeomorphism of their regular neighborhoods and, because the two complementary components are disks, extends over all of the surface. This proves the usual uniqueness of an ordered five-chain on a genus-2 surface without appealing to a picture.

The involution reverses the chain, exchanging $a\leftrightarrow e$ and $b\leftrightarrow d$ and preserving c; the half-turn has the same action on the polygonal ribbon graph. Choose the graph isomorphism equivariantly on one curve and its image at each stage. Both complementary disks are invariant: if they were exchanged, neither could contain a fixed point, whereas φ has two fixed points and none lies on the chain. Brouwer's fixed-point theorem gives a fixed point in each invariant disk, hence exactly one in each. To extend equivariantly, quotient each complementary disk by its order-two action. The quotient is a disk with one branch point. The equivariant boundary identification descends to a boundary diffeomorphism of the quotient disks. Extend it over the disks while sending branch point to branch point, choosing the extension to be linear in oriented branch charts, and lift it using the boundary-determined lift. The linear local form makes the lift smooth at the fixed point, and the lifted diffeomorphism conjugates the two disk actions. Doing this on both components sends the fixed points to p, O and completes the equivariant marked identification. □

The auxiliary curve z from [LP25, Figure 1] is deliberately not drawn in Figure 1: it is disjoint from c, whereas a previously used symmetric two-chord representative was not. Only its source-figure intersection data and the resulting homology class $[z] = [b] - [d]$ are used in the diagnostic calculations of Part I; neither z nor a based word for it is used in the proof of Theorem D.

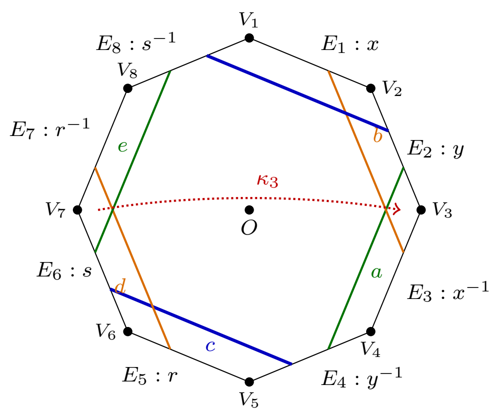

FIGURE 1. The fiber F: the regular octagon with edge word $xyx^{-1}y^{-1}rsr^{-1}s^{-1}$, all eight vertices identified to the single basepoint p, so that each edge is a based loop. The rotation by π induces the involution φ_0 ($x \leftrightarrow r$, $y \leftrightarrow s$, exactly), with fixed points p and the center O. The solid curves form the Lidman–Piccirillo five-chain a, b, c, d, e. Identified endpoints on paired edges close each curve. The dotted path lies in the middle component of the octagon cut along the two blue c-arcs and represents κ_3; it is included here because its disjointness from c is used in Section 10.

7.2. **Curves and lifts.** The twist curves are the displayed a, b (push offs of the x- and y-circles, crossing the y- resp. x-circle once near p) and their ρ-images $e := \varphi_0(a)$, $d := \varphi_0(b)$, so Lidman–Piccirillo's symmetry (φ: $a \leftrightarrow e$, $b \leftrightarrow d$) is built into the model by Lemma 7.1. The twist directions are normalized so that the based actions are $T_a\colon (x, y) \mapsto (x, yx)$, $T_b\colon (x, y) \mapsto (xy^{-1}, y)$; a twist direction is a binary choice per curve, some choice realizes these standard once-punctured-torus formulas, and the residual chirality ambiguity is absorbed by the handle swap (Remark 3.3); the composite $h := T_a \circ T_b\colon x \mapsto y^{-1},\ y \mapsto yx$ is the trefoil monodromy of Lemma 3.2, which fixes the boundary word $[x, y]$ exactly and satisfies $h^6 = \mathrm{conj}_{[x,y]^{-1}}$ (the boundary Dehn twist) and $h^3 = -\mathrm{id}$ on H_1. Set $\tilde{\psi} := h * \mathrm{id}$

(well defined on the closed-fiber group because h fixes $[x, y]$ exactly) and $\tilde{\varphi} :=$ the swap; the identity $\tilde{\psi}\tilde{\varphi}\tilde{\psi}^{-1}\tilde{\varphi}^{-1} = h * h^{-1}$ of Lemma 3.4(2) realizes Lidman–Piccirillo's factorization $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$. The diffeomorphism $\psi_0 := T_a T_b$ fixes p, O and the entire right handle pointwise; in particular $\psi_0(e) = e$ pointwise.

The invariant curve used below is the two-arc curve fixed by the marked identification: $c = \gamma \cup \rho(\gamma)$, with γ joining the y-edge pair to the s-edge pair. Its embedded isotopy class comes from Lemma 7.1; its *based words*, which also depend on orientation and basing, are derived in Section 8.

7.3. **The cut-square model of the bundle.**

Convention 7.2 (Words, composition, and the base generators)**.** Paths and loops compose *left to right*: $\gamma_1\gamma_2$ traverses γ_1 first. Relators are words equal to 1; a relation written $ugu^{-1} = w$ means the corresponding relator $ugu^{-1}w^{-1}$. The base generators are $A := [\bar{\alpha}]^{-1}$ and $B := [\bar{\beta}]^{-1}$, so that conjugation by B implements one *upward* transport around $\bar{\beta}$: $BgB^{-1} = \tilde{\psi}(g) \cdot [\text{corr}]$ for a fiber loop g, and likewise A for $\bar{\alpha}$ and $\tilde{\varphi}$. (The opposite choices are the ε-flips of Remark 3.7; they are covered by the sign checks of Remark 11.1, but all displayed words in this part use the stated convention.)

The base T_0 (once-punctured torus) is the square $[0, 1]^2$ minus a puncture disk at $(3/4, 3/4)$, with $(\xi, (t, 1)) \sim (\psi_0\xi, (t, 0))$ and $(\xi, (1, u)) \sim (\varphi_0\xi, (0, u))$: crossing the α-cut upward applies ψ_0; crossing the β-cut rightward applies φ_0. Basepoint (p, q) with $q = (1/4, 1/4)$; based base loops $\bar{\alpha}$ (horizontal through q; holonomy $\tilde{\varphi}$) and $\bar{\beta}$ (vertical; holonomy $\tilde{\psi}$), realizing the minimal crossing pattern with the embedded curves: $\bar{\alpha}$ meets the embedded β once and the embedded α not at all, and symmetrically for $\bar{\beta}$, minimal because $|\alpha \cdot \beta| = 1$ forces one crossing, while T_0 cut along α still contains a boundary-parallel copy of α basable at q with trivial basing correction. The surgery tori sit on the cuts: $T_\alpha = c \times \{\alpha\text{-cut}\}$ (closing because $\varphi_0(c) = c$; the restriction $\varphi_0|_c$ is a free half-rotation, a framing fact used in Section 8.4) and $T_\beta = e \times \{\beta\text{-cut}\}$ (closing with the product framing because $\psi_0(e) = e$ pointwise). Figure 2 shows the base square and the domain of a transport annulus.

We will show that of the eight monodromy relations (four fiber generators over each of the two base loops), *exactly three acquire meridian corrections*: s over $\bar{\alpha}$, and y and s over $\bar{\beta}$, each from a single transverse crossing; the other five stay clean. (Three, not the four of the embedded count of Section 6.2: Remark 6.1 reconciles the two.) The mechanism, one relation at a time:

Each monodromy relation of the bundle is realized by an annulus, which we now make explicit, because its intersections with the surgery tori are the whole story. For a fiber generator g transported around $\bar{\beta}$, consider the map of a square

$$\mathfrak{M}_g(u, v) = \big(\text{transport of } g(u) \text{ along } \bar{\beta} \text{ to time } v\big) :$$

its bottom edge is g, its top edge is the transported loop $\tilde{\psi}(g)$, and both vertical edges are $\bar{\beta}$ itself (the track of the basepoint, since the monodromies fix p); as the two vertical edges coincide, the image is an annulus in the manifold, the *transport annulus* of g. In the unsurgered bundle R this annulus *is* the homotopy behind the monodromy relation $BgB^{-1} = \tilde{\psi}(g)$. In the surgered manifold the annulus may meet the surgery tori, and each transverse intersection point punctures it: the relation then holds only up to one conjugated-meridian factor per puncture, $BgB^{-1} = (\ell\,\mu^{\pm 1}\ell^{-1}) \cdots \tilde{\psi}(g)$, where ℓ, the *conjugating path* of the correction, runs in the annulus from the base corner to the puncture (Section 8.3 turns each such factor into an explicit word). Because each surgery torus is a product (fiber

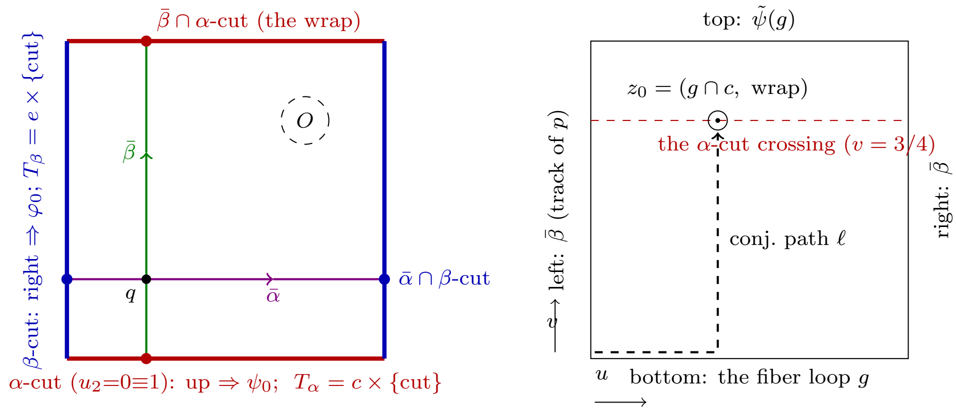

FIGURE 2. Left: the cut square, its basepoint $q = (1/4, 1/4)$, the based loops $\bar\alpha, \bar\beta$, the puncture O, and the base circles carrying the surgery tori. All T_α-corrections happen over the single marked point $\bar\beta \cap \alpha$-cut, all T_β-corrections over $\bar\alpha \cap \beta$-cut. Right: the domain of the transport annulus $\mathfrak{M}_g$ for g transported around $\bar\beta$; it meets T_α only on the dashed line (the α-cut crossing), and only at fiber points of $g \cap c$: for $g = y$ the single puncture z_0, whose conjugating path ℓ is shown. (Every crossing count used in the derivation is also verified mechanically; Appendix A.)

curve) $\times$ (base cut circle), the punctures are exactly the points (base crossing of the loop with the cut) $\times$ (fiber crossing of g with the torus' fiber curve): a finite count, read off the intersection data of Section 8.2.

Remark 7.3 (A worked example: the corrected relation for y)**.** Take $g = y$ over $\bar\beta$, the simplest corrected relation. The base track of the transport annulus is the vertical circle through $q = (1/4, 1/4)$: traveling upward from q, it crosses the α-cut exactly once, at the point $(1/4,\ 1{\equiv}0)$ where the vertical circle wraps around (*the wrap*, for short), and never meets the β-cut (its column stays at $u_1 = 1/4$). So this annulus can meet only $T_\alpha = c \times \{\alpha\text{-cut}\}$, and it does so exactly where its fiber point lies on c: the loop y crosses c once, at c_y (Section 8.2), so the annulus meets T_α in the single transverse point $(c_y,\ (1/4, 1{\equiv}0))$. Puncturing there, the clean relation $ByB^{-1} = \tilde\psi(y) = yx$ acquires exactly one conjugated-meridian factor, whose conjugating path runs in the annulus to the puncture: the partial loop y_1 (from p to c_y) followed by the vertical transport up to the cut. Defining the based meridian M *along that very arc* ($M := y_1 \cdot (\text{meridian circle at } c_y) \cdot y_1^{-1}$, Section 8.3) absorbs the conjugating path entirely, and the corrected relation is $ByB^{-1} = M^{\pm 1}(yx)$. Every other transport annulus is this computation with different intersection data. Over $\bar\beta$: the loop s crosses c once, at the point c_s (Section 8.2), giving $BsB^{-1} = (\delta M^{\pm 1}\delta^{-1})s$ with the explicit conjugating path δ (the meridian is based at c_y, so the basing must slide along c from c_s: that slide is δ); the loops x and r miss c entirely, so their relations stay clean. Over $\bar\alpha$: the base track (the horizontal circle through q) crosses the β-cut once and never meets the α-cut, so only $T_\beta = e \times \{\beta\text{-cut}\}$ is in play; the loop s crosses e once at s_e, giving $AsA^{-1} = N^{\pm 1}y$ (with N based along the arc s_2, again absorbing its own conjugating path), while x, y, r miss e, so their relations stay clean. That is the full count: three corrected relations, five clean ones, each correction at a single crossing point. The same point-versus-circle discipline governs the *push off* basings: there the object transported around $\bar{\beta}$ is a basing arc rather than a generator, and the square it sweeps is subject to the same crossing count (Section 8.5).

8. The derived words

8.1. **The invariant curves.** The curve c (it crosses E_8, then E_4) has based word $(rx)^{-1}$ when based at the V_2 corner; developing c at the V_1, V_2, V_3 and V_5 basings (Appendix A) realizes the xr/rx ordering ambiguity as a pure choice of basing. The auxiliary source curve z is not part of this polygonal word calculation and is not needed below; its corrected homology class is recorded in Section 3.2.

8.2. **Intersection data.** Disjointness pins the normal positions: on the s-edge, the c-crossing precedes the e-crossing (else $c \cap e \neq \varnothing$); on the y-edge only the c-crossing is relevant. We name the three points that recur: c_y and c_s are the single crossings of c with the y- and s-edges, and s_e is the single crossing of e with the s-edge; on the s-edge, c_s comes before s_e. Realizability of the five-chain in the frozen polygonal coordinates is checked mechanically (Appendix A): c meets the strip of d (the band between a push-off curve and its edge circle) on leaving E_6 and the strip of b on leaving E_2; all five curves avoid p and O.

8.3. **Meridians, conjugating paths, and the corrected relations.** Let y_1 and s_1 be the initial segments of the based loops y and s up to their c-crossings, and s_2 the initial segment of s up to its e-crossing (on the s-edge the c-crossing comes first, by the intersection data). Base the meridians at the surgery-relevant crossings: $M := y_1 \cdot (\text{meridian circle of } T_\alpha) \cdot y_1^{-1}$, $N := s_2 \cdot (\text{meridian circle of } T_\beta) \cdot s_2^{-1}$. Then the transport-annulus analysis gives the three corrected relations in the form

$$ByB^{-1} = M^{\pm}\,(yx), \qquad BsB^{-1} = \left(\delta M^{\varepsilon}\delta^{-1}\right) s, \qquad AsA^{-1} = N^{\pm}\, y,$$

i.e. two of the three corrections are *exactly* meridian powers (the basing absorbs the conjugating path), and the third has conjugating path $\delta = s_1 \cdot (\text{arc of } c) \cdot y_1^{-1}$, computed mechanically (Appendix A): $\delta = r^{-1}$ (via one arc of c) or $\delta' = x$ (via the other), with the arc-difference identity $\delta'\delta^{-1} = xr \sim c$ (validation V4, Appendix A). Both values of every sign are checked ($\varepsilon \in \{\pm 1\}$ denotes the sign of the Bs-correction, which reappears in Section 8.5); the left placement of the corrections is tied to the stated annulus orientation and is probed by a full placement check (Remark 11.1). The crossing $z_0 = (c_s, \text{the wrap of } \bar{\beta})$ that produces the Bs-correction corrects one more word: the base-direction push off of T_β (Section 8.5).

8.4. **The direction (Lagrangian-framing) words.** For a loop on a surgery torus traveling once in the base direction and crossing one cut with holonomy Θ at fiber position η, the based push off has the form

$$(\text{cut generator}) \cdot \tilde{\Theta}(\gamma_\eta) \cdot (\text{drift}) \cdot \gamma_\eta^{-1},$$

where γ_η is a fiber path from p to η and the *drift* is the fiber path along which the holonomy carries the closing curve: trivial for T_β, where ψ_0 fixes e pointwise, and half of c for T_α, as derived below. This yields

$$\mathrm{dir}^{\mathrm{base}}_{T_\beta} = \left(\delta M^{-\varepsilon}\delta^{-1}\right) B, \qquad \mathrm{dir}^{\mathrm{fib}}_{T_\beta} = s\, r^{-1} s^{-1},$$
$$\mathrm{dir}^{\mathrm{base}}_{T_\alpha} = A\, r^{-1}, \qquad \mathrm{dir}^{\mathrm{fib}}_{T_\alpha} = (rx)^{-1}.$$

Line by line. On T_β the drift is zero, and the meridian conjugate in the base direction is the push-off basing correction derived in Section 8.5; the fiber direction is e based along s_2;

writing e_{V_7} for the development of e departing from the wedge V_7 (Appendix A), one has $s\,r^{-1}s^{-1} = s \cdot e_{V_7} \cdot s^{-1}$, as it must be. On T_α the half-rotation drift is *exactly* the conjugating path: $\varphi_0(c_y) = c_s$ says the half-rotation carries one crossing of c to the other (Section 8.2), and together with $\tilde{\varphi}(y_1) = s_1$ this is why the drift equals the conjugating path; the fiber direction is c at the y_1-basing. Because $\varphi_0|_c$ is a free half-rotation, the base direction on T_α cannot close without drifting along half of c; we write λ for the resulting closed curve, the base-direction push off itself. The identity $\lambda^2 = x^{-1}A^2r^{-1}$ (on exponent sums, $2\lambda = 2[A] - [c]$, since $[c] = [x] + [r]$) holds exactly and is the sharpest internal check on this, the single most delicate input word. The (m,n)-family directions are $\mathrm{dir}^{\mathrm{base}} \cdot (\mathrm{dir}^{\mathrm{fib}})^{m \text{ or } n}$; the two arc conventions differ by exactly one unit of n, hence are absorbed by the n-range of the family.

8.5. **The push-off basing correction.** The push-off formula above is a bundle identity; realizing it in the *complement* of the surgery tori transports the basing arc s_2 once around $\bar{\beta}$, which sweeps the square

$$\mathfrak{A}(u,v) = \big(\text{transport of } s_2(u) \text{ along } \bar{\beta} \text{ to time } v\big),$$

a square whose sides are s_2, the push off's base circle at fiber s_e, $\tilde{\psi}(s_2) = s_2$ (pointwise: ψ_0 fixes the right handle), and $\bar{\beta}$.

The square misses T_β: its base column never meets the β-cut, and its interior fiber points $s_2(u)$, $u < u_e$ (u_e the endpoint parameter of s_2), miss e (intersection data: one e-crossing on the s-edge, at the endpoint of s_2). It crosses T_α at exactly one interior point $z_0 = (c_s, \text{the wrap of } \bar{\beta})$: the base circle crosses the α-cut once, and s_2 crosses c exactly once, at c_s; the intersection data's own ordering (the c-crossing precedes the e-crossing on the s-edge) places that crossing *inside* s_2. The local intersection number is ± 1, and the crossing count is homological: the single crossing is interior, so $\partial\mathfrak{A}$ misses T_α and $\mathfrak{A}$ defines a class in $H_2\big(R,\, R \smallsetminus \nu(T_\alpha)\big) \cong \mathbb{Z}$, whose pairing with the Thom class of $\nu(T_\alpha)$ counts transverse crossings with sign and depends on $\mathfrak{A}$ only through its homotopy class rel boundary. No interior homotopy avoids the crossing (the disjointness $c \cap e = \varnothing$ protects only the constant-fiber slide of the push off circle, not the transported basing arc; see Remark 8.1).

Puncturing the swept disk at z_0 and reducing by the *same* steps that produced the Bs-correction (the crossing point, the conjugating path and the arc route are literally those of Section 8.3, because $\mathfrak{A}$ is the restriction of the transport annulus of s to $u \le u_e$) gives, with ε the sign of the Bs-correction in the same convention,

$$\mathrm{dir}^{\mathrm{base}}_{T_\beta} = \big(\delta M^{-\varepsilon} \delta^{-1}\big)\, B, \qquad \delta = r^{-1} :$$

the B-conjugation in the transport identity cancels exactly (so the insertion is on the left, with the plain conjugating path δ), and the meridian sign is *anti-coupled* to the Bs-correction sign. Both are relative statements, hence covariant under the global orientation flip that the sign checks of Remark 11.1 already cover; no new convention is introduced. The correction abelianizes to zero, so the homology bookkeeping of Lemma 4.2 is unchanged; and the arc choice is immaterial in the group, since $\delta' M \delta'^{-1}$ and $\delta M \delta^{-1}$ differ by conjugating M by a based copy of c (the V4 identity), a push off class on the boundary 3-torus $\partial\nu(T_\alpha)$, which commutes with the meridian there.

Remark 8.1 (The history of this word)**.** This is the one input word that an earlier version of this derivation got wrong: that version asserted that the swept square misses T_α because $c \cap e = \varnothing$, a disjointness that protects the constant-fiber slide of the push off circle but not the transported basing arc. Testing a circle's transport by the crossings of a single point is precisely the kind of error the T^4 calibration detects, and it is what caught this one (Appendix B.1). With the corrected, sign-coupled word every conclusion was re-established: the grid of sign assignments, the placement probe, the second diagram, and the Knuth–Bendix checks were all rerun, and all nine formal exponent systems define trivial relation groups in every convention (Section 11); Theorem D uses only $(0,0)$. Before the derivation was pinned down, all eight candidate resolutions of the correction (both arcs, both signs, both placements) were checked at the surgery variant $(0,0)$ (256 systems, each decided by enumeration or rewriting). Thus simple connectivity at the specified variant does not depend on resolving that local sign-and-placement choice, although the displayed relation still records the geometric derivation.

8.6. **Two known-answer calibrations.** The method of this section (the cut-model conventions, the crossing counts of the transport annuli, the meridian basings and conjugating paths, the surgery-relation format, and the finite-group decision procedures) was tested on two configurations whose answers are known independently; full accounts are in Appendix B.

First, the disjoint Lagrangian tori $T_1 = X \times A_1$, $T_2 = Y \times A_2$ in T^4 of Baldridge–Kirk [BK09], whose complement presentation they computed with explicit care about basings: double ± 1 Luttinger surgery gives $\pi_1 = (\mathbb{Z}^2 \rtimes_A \mathbb{Z}) \times \mathbb{Z}$ with $|\operatorname{tr} A| \in \{1, 3\}$ according to the relative signs, non-abelian in every convention. Deriving the input words for this configuration by the same transport-annulus method reproduces the ground truth in every sign and placement convention, while every wrong candidate for the basing data fails; it is this test that caught the one wrong input word of the first version of this derivation (Remark 8.1).

Second, a Seifert-fibered configuration exercising exactly the machinery the T^4 test cannot (the T^4 has trivial monodromy): the closed manifold $N \times S^1$, N the mapping torus of φ_0, surgered along the base-direction push-off curve λ times the extra circle. There the surgeries are torus twists, the resulting graph-manifold groups are classified, and the derived words (the conjugating path $\delta = r^{-1}$, the anti-coupled meridian sign, the push-off label) reproduce the classified groups at three surgery coefficients while seven families of wrong words fail at every coefficient. Together the two tests exercise every input class of the derivation: meridian basings, corrections, transport laws, and the base-direction push offs of both tori.

8.7. **The framing lemma.** Luttinger surgery is performed with respect to the *Lagrangian framing*: in a Weinstein chart (a symplectomorphism of $\nu(T)$ with a neighborhood of the zero section of T^*T^2, canonical coordinates $(\Theta_1, \Theta_2; P_1, P_2)$, $\omega = dP_1 \wedge d\Theta_1 + dP_2 \wedge d\Theta_2$, $T = \{P = 0\}$) the framing push off of a curve on T is its lift at *constant momentum*. Two framings of T differ by adding meridian multiples to the push-off classes, and a meridian error in a direction word changes the surgery coefficient, and no check over sign conventions can protect against one. The following lemma closes this, the most delicate input.

Lemma 8.2 (Framing)**.** *Fix the Thurston-type symplectic form on R built from a fiber area form invariant under both monodromies, chosen in the normalized local models below. This is precisely the class of forms used in* [LP25, Section 1]*: their form on R is the one induced by "an area form on the fiber preserved by both monodromies", with no further constraint, so the normalized choice below is one of theirs. Then for both T_β and T_α the fibered push offs used in Section 8.4 (the in-fiber parallel copy of the fiber curve, and the base-direction shift of the closed base-direction curve) are constant-momentum lifts in an explicit Weinstein chart.*

In particular their classes on $\partial\nu(T)$ have zero meridian component: the fibered framing is *the Lagrangian framing.*

Proof. T_β*.* The twist supports of ψ_0 lie in the left handle, so $\psi_0 = \mathrm{id}$ on a neighborhood of e; hence the bundle is a genuine product over a base annulus near the β-cut. Normalize (Moser, on an annulus) the invariant fiber area form to $dt \wedge d\theta_1$ on $A_e \cong S^1 \times (-1,1)$ and write the base annulus as (θ_2, u). The Thurston-type form on this chart is $\omega = dt \wedge d\theta_1 + K\,du \wedge d\theta_2$, which is canonical for $(\theta_1, \theta_2;\ t,\ Ku)$, with $T_\beta = \{t = u = 0\}$ the zero section. The fiber push off $\{t = t_0, u = 0\}$ and base push off $\{t = 0, u = u_0\}$ (t_0, u_0 small constants) sit at constant momentum. T_α*.* Here the holonomy around the α-cut is φ_0 with $\varphi_0|_{A_c}$ an area-preserving free half-rotation; normalize it (equivariant Moser) to $(\theta_1, t) \mapsto (\theta_1 + \pi, t)$ with fiber form $dt \wedge d\theta_1$. The ν-region is the quotient of $A_c \times [0, 2\pi] \times (-1,1)$ by $(\theta_1, t, 2\pi, u) \sim (\theta_1 + \pi, t, 0, u)$, with $\omega = dt \wedge d\theta_1 + K\,du \wedge d\theta_2$ descending to it. The change of coordinates

$$\Theta_1 = \theta_1 - \tfrac{\theta_2}{2}, \qquad \Theta_2 = \theta_2, \qquad P_1 = t, \qquad P_2 = \tfrac{t}{2} + Ku$$

is well defined on the quotient (the deck identification sends $\Theta_1 \mapsto \Theta_1 + 2\pi$) and gives $\omega = dP_1 \wedge d\Theta_1 + dP_2 \wedge d\Theta_2$ with $T_\alpha = \{P = 0\}$: an explicit Weinstein chart on the mapping torus of the half-rotation. Our fiber push off $\{t = t_0, u = 0\}$ has $P = (t_0, t_0/2)$; our base push off $\{t = 0, u = u_0\}$, whose base-direction closed curve is precisely the drift curve λ ($\Theta_1 = \mathrm{const}$, i.e. once around the base while drifting half-way along c), has $P = (0, Ku_0)$. Both are constant-momentum lifts. The Lagrangian framing is independent of the choice of Weinstein chart [ADK03], and the based words of Section 8.4 are the developments (Appendix A) of exactly these push-off curves.

It remains to supply the two normalizations invoked. *Moser on the annulus.* Write the invariant fiber area form near e as $\Omega = f\,dt \wedge d\theta_1$ on $A_e \cong S^1 \times (-1,1)$, $f > 0$, $e = \{t = 0\}$, and set $\Omega_0 = dt \wedge d\theta_1$. Put $g(\theta_1, t) = \int_0^t \big(f(\theta_1, \tau) - 1\big)\,d\tau$ and $\zeta = g\,d\theta_1$: then $d\zeta = \Omega - \Omega_0$ and ζ vanishes identically along e. Each $\omega_s = (1-s)\Omega_0 + s\Omega$ is an area form (a convex combination of positive multiples of $dt \wedge d\theta_1$), so there is a unique vector field X_s with $\iota_{X_s}\omega_s = -\zeta$; it vanishes along e, so its flow ϕ_s exists for $s \in [0,1]$ on a neighborhood of the compact core and fixes e pointwise, and $\frac{d}{ds}\phi_s^*\omega_s = \phi_s^*\big(d\,\iota_{X_s}\omega_s + d\zeta\big) = 0$. Hence $\phi_1^*\Omega = \Omega_0$: near e there is a chart in which the invariant form *is* $dt \wedge d\theta_1$. Only the germ of the chart near the surgery torus enters the constructions above, so this suffices. *Equivariant Moser for the half-rotation.* On a collar A_c of c the involution $\varphi_0|_{A_c}$ is free ($\mathrm{Fix}(\varphi_0) = \{p, O\}$ misses the collar), orientation-preserving, preserves c with its orientation (hence preserves each side of c) and preserves Ω. The quotient $\bar{A} = A_c/\varphi_0$ is therefore again an annulus with core $\bar{c} = c/\varphi_0$, the quotient map π is a connected double cover, and Ω descends: $\Omega = \pi^*\bar{\Omega}$. Apply the previous paragraph downstairs: a diffeomorphism $\bar{\psi}$ of a collar of $\bar{c}$, fixing $\bar{c}$ pointwise, with $\bar{\psi}^*\bar{\Omega} = dt \wedge d\bar{\theta}$. Fixing $\bar{c}$ pointwise, $\bar{\psi}$ induces the identity on π_1, so it lifts to a diffeomorphism ψ of the corresponding collar in A_c; the conjugate $\psi^{-1}(\varphi_0|)\psi$ is a deck transformation covering the identity and is not the identity, hence equals $\varphi_0|$: the lift is equivariant. In the lifted coordinates the double cover of $\big(S^1_{\bar{\theta}} \times (-1,1),\ dt \wedge d\bar{\theta}\big)$ is $S^1_{\theta_1} \times (-1,1)$ with deck transformation $(\theta_1, t) \mapsto (\theta_1 + \pi, t)$ and pulled-back form $2\,dt \wedge d\theta_1$; absorbing the factor 2 into t gives exactly the normalized model used above. □

Remark 8.3. The chart re-derives the drift from the symplectic side: in the coordinates above, the closed curves on T_α are generated by the Θ_1-circle ($= c$) and the Θ_2-circle ($= \lambda$); a "pure base" curve does not exist on this torus, so the half-drift in $\mathrm{dir}^{\mathrm{base}}_{T_\alpha} = Ar^{-1}$ is forced twice over: topologically (Section 8.4) and symplectically. The lemma is stated for the exhibited normalized form; [LP25]'s construction permits this choice, all downstream arguments hold for it, and the surgery coefficient's sign convention remains among the conventions checked in Remark 11.1.

9. The presentation

Generators x, y, r, s (fiber), A, B (base), M, N (meridians). Relations:

(i) $[x, y][r, s]$;
(ii) clean $\tilde{\varphi}$-monodromy: $AxA^{-1} = r$, $AyA^{-1} = s$, $ArA^{-1} = x$;
(iii) clean $\tilde{\psi}$-monodromy: $BxB^{-1} = y^{-1}$, $BrB^{-1} = r$;
(iv) corrected monodromy, as in Section 8.3 (signs as in (C7));
(v) fillings: $M \cdot (Ar^{-1} \cdot ((rx)^{-1})^n)^{\pm 1}$, $N \cdot \big((\delta M^{-\varepsilon}\delta^{-1})B \cdot (sr^{-1}s^{-1})^m\big)^{\pm 1}$ ($\varepsilon =$ the sign of the Bs-correction in (iv), Section 8.5);
(vi) the drilled-fiber relation R_3 of Section 10: $B\,(s^{-1}r^{-1}yx)\,B^{-1} = r^{-1}s^{-1}x$, a relation true by the geometric derivation there.

As checks: dropping (v) gives $H_1 = \mathbb{Z}^2$; fiber-only fillings give $H_1 = \mathbb{Z}^2$; one filling gives $H_1 = \mathbb{Z}$; the full relation system has $H_1 = 0$ for every tested formal exponent choice and sign, exactly as the homology analysis of Lemma 4.2 requires.

9.1. **The assembled presentation at the Lidman–Piccirillo surgery.** Because the relations above are stated schematically, with (iv) referring back to Section 8.3 and δ and the signs of (C7) left as parameters, we write the presentation out in full at the variant that matters, $(m, n) = (0, 0)$. Table 1 is the object against which an independently derived presentation should be compared; the protocol for doing so is Section A.4. Generators: fiber x, y, r, s (octagon edge loops of F, based at p; y is [LP25]'s curve class b and s the class d), base A, B (C3), and meridians M of T_α based along y_1, N of T_β based along s_2 (Section 8.3).

Two words enter only away from $(0, 0)$, as the m-th and n-th powers appended in (v): $\mathrm{dir}^{\mathrm{fib}}_{T_\alpha} = (rx)^{-1}$ (the derived word of c) and $\mathrm{dir}^{\mathrm{fib}}_{T_\beta} = s\,r^{-1}s^{-1}$ (the s-based word of e).

At this variant the outcome is: $H_1 = 0$ in all conventions; coset enumeration gives $|G| = 1$ in all 32 sign assignments of the relation system; and Knuth–Bendix reduces every generator to 1 in all conventions and in both based diagrams (Section 11). Dropping F1 and F2 gives $H_1 = \mathbb{Z}^2$; dropping one of them gives $H_1 = \mathbb{Z}$.

10. A surjective relation system

For the simple-connectivity theorem it is unnecessary to prove that the relations of Section 9 present the fundamental group exactly. It is enough to prove that they hold geometrically and that their generators generate the true group: the group defined by the displayed relations then surjects onto the true group, and triviality descends through that surjection. One additional clean relation materially improves the group decision. We first derive it from an explicit drilled-fiber basis.

Lemma 10.1 (Drilled-fiber bases)**.** *With the curves and basings of Figure 1,*

$$\pi_1(F \setminus \nu e, p) = F_3\langle x, y, r\rangle, \qquad \pi_1(F \setminus \nu c, p) = F_3\langle x, r, \kappa_3\rangle,$$

where κ_3 is the dotted middle-region loop in the figure and

$$\kappa_3 = s^{-1}r^{-1}yx \quad \textit{in } \pi_1(F, p).$$

In particular κ_3 has a representative geometrically disjoint from c.

#	relator	derivation and machine check
R0	$[x,y]\,[r,s]$	the genus-2 octagon surface relator (Section 7.1)
A1	$A\,x\,A^{-1} = r$	clean $\tilde{\varphi}$-monodromy (the swap); lift machine-checked (Appendix A)
A2	$A\,y\,A^{-1} = s$	same
A3	$A\,r\,A^{-1} = x$	same
B1	$B\,x\,B^{-1} = y^{-1}$	clean $\tilde{\psi}$-monodromy ($h\colon x \mapsto y^{-1}$); machine-checked (Appendix A)
B2	$B\,r\,B^{-1} = r$	clean $\tilde{\psi}$-monodromy ($h * \mathrm{id}$ fixes the right handle pointwise)
R3	$B\,(s^{-1}r^{-1}yx)\,B^{-1} = r^{-1}s^{-1}x$	the drilled-fiber relation $B\kappa_3B^{-1} = \tilde{\psi}(\kappa_3)$, κ_3 the third basis element of $\pi_1(F \smallsetminus \nu c)$; development record D5, Appendix A; Section 10
M1	$A\,s\,A^{-1} = N^{\varepsilon_3}\,y$	the transport annulus $s \times \bar{\alpha}$ crosses T_β once, at (s_e, β-cut): $\bar{\alpha}$ meets the β-cut once and $s \cap e = \{s_e\}$ (intersection data, record D4); N's s_2-basing absorbs the conjugating path. Sections 7.3, 8.3
M2	$B\,y\,B^{-1} = M^{\varepsilon_4}\,(yx)$	the transport annulus $y \times \bar{\beta}$ crosses T_α once, at (c_y, α-cut wrap); M's y_1-basing absorbs the conjugating path. Remark 7.3 works this case end to end
M3	$B\,s\,B^{-1} = \delta\,M^{\varepsilon}\delta^{-1}\,s,\quad \delta = r^{-1}$	the transport annulus $s \times \bar{\beta}$ crosses T_α once, at $z_0 = (c_s, \mathrm{wrap})$; the conjugating path δ is the slide of the meridian basing along c from c_s to c_y (development record D3; V4 arc identity). Section 8.3
F1	$M\,(A\,r^{-1})^{\varepsilon_A} = 1$	T_α filling, direction $\alpha' c^n$ at $n = 0$: $\mathrm{dir}^{\mathrm{base}}_{T_\alpha} = Ar^{-1}$, the half-rotation drift being exactly the conjugating path ($\varphi_0(c_y) = c_s$, $\tilde{\varphi}(y_1) = s_1$); framing Lemma 8.2; drift identity $2\lambda = 2[A] - [c]$ exact. Section 8.4
F2	$N\,\big((r^{-1}M^{-\varepsilon}r)\,B\big)^{\varepsilon_B} = 1$	T_β filling, direction $\beta' e^m$ at $m = 0$: $\mathrm{dir}^{\mathrm{base}}_{T_\beta} = (\delta M^{-\varepsilon}\delta^{-1})B$: the push-off basing correction, meridian sign anti-coupled to ε; found by the T^4 calibration. Section 8.5; Appendix B.1

TABLE 1. The relation system for V at the Lidman–Piccirillo variant $(m,n) = (0,0)$, one line per relation. Signs are as in (C7); the conclusion holds for every assignment.

Proof. Remove open collars of the displayed chords before making the edge identifications. The single e-chord cuts the octagon into two disks. In each disk contract a tree joining its edge intervals and the appropriate copy of the e-collar. After the paired polygon edges are identified, the three uncontracted edges are the loops x, y, r; the resulting rose is a spine of $F \setminus \nu e$. Equivalently, the quotient cell structure has $V = 3, E = 7, F = 2$ and $\chi = -2$, while the explicit tree collapse leaves three loops.

The two nonintersecting c-chords cut the octagon into three disks: the upper, middle and lower regions visible in Figure 1. Contract in each region a tree joining all its polygon-edge intervals and collar intervals, retaining the x-edge pair, the r-edge pair, and one arc across the middle region from the V_7 wedge to the V_3 wedge. The quotient is a three-petal rose, represented by x, r, κ_3, and is a spine of $F \setminus \nu c$. This time the unreduced cell count is $V = 5, E = 10, F = 3$, again $\chi = -2$. The retained middle arc is disjoint from both c-chords by inspection of the displayed polygon; the same strict segment-disjointness is checked from the frozen endpoint coordinates by record D5 in Appendix A.1. Developing it from V_7 to V_3 without crossing a polygon edge gives $s^{-1}r^{-1}yx$. Thus the collapse proves both the geometric disjointness and the asserted free basis; the vanishing of algebraic intersection is only a secondary check. $\square$

The first basis shows that (ii)–(iii) already contain the full clean set for T_β: no completion is needed. For T_α, the clean relations in (iii) cover x, r and Lemma 10.1 supplies exactly one further basis element. Its transport annulus around B misses T_α because $\kappa_3 \cap c = \varnothing$, and it misses T_β because the based loop $\bar{\beta}$ is disjoint from the β-cut. Hence it gives the true clean relation

$$R_3 : \qquad B\,\kappa_3\,B^{-1} = \tilde{\psi}(\kappa_3) = r^{-1}s^{-1}x.$$

For the last equality, $\tilde{\psi}(\kappa_3) = s^{-1}r^{-1}yxy^{-1}$ and the surface relation gives $yxy^{-1} = [r, s]x$, so $s^{-1}r^{-1}yxy^{-1} = r^{-1}s^{-1}x$. Thus R_3 is a relation in the actual complement, not an algebraically disjoint surrogate. Adjoining it passes to a quotient of the relation group, so earlier trivial outcomes remain trivial.

Lemma 10.2 (Surjection onto the manifold group)**.** *For fixed choices of the signs in (C7), let G be the group on x, y, r, s, A, B, M, N defined by relations* (i)–(vi) *of Section 9 at $(m, n) = (0, 0)$. There is a natural epimorphism*

$$G \twoheadrightarrow \pi_1(V).$$

Consequently, $G = 1$ implies $\pi_1(V) = 1$.

Proof. Put $C = R \setminus \big(\nu T_\alpha \cup \nu T_\beta\big)$. General position makes every loop in R disjoint from the two codimension-two tori, so $\pi_1(C) \to \pi_1(R)$ is surjective. Moreover, its kernel is normally generated by meridians of the two tori: make a null-homotopy in R transverse to the tori and remove small disks around its intersection points; the new boundary circles are conjugates of the meridians. Since x, y, r, s, A, B generate $\pi_1(R)$, chosen lifts of those loops together with the based meridians M, N therefore generate $\pi_1(C)$.

The inclusion $C \hookrightarrow V$ is surjective on fundamental groups by van Kampen; each attached $T^2 \times D^2$ kills its stated filling slope. Relations (i)–(iii) hold by the surface and clean transport annuli, the corrected relations (iv) by Section 8.3, the filling relations (v) by Sections 8.4–8.7, and relation (vi) by the drilled-fiber argument immediately above. Hence the map from the free group on the eight displayed generators to $\pi_1(V)$ kills every defining relation of G and factors through the asserted epimorphism. $\square$

We do not assert that this epimorphism is an isomorphism, and no such assertion is needed below. A nontrivial computation of G would be inconclusive; the trivial computation below passes to every quotient.

11. Deciding the relation groups

Let G denote the geometrically derived group of Lemma 10.2. By that lemma, it is enough to prove $G = 1$. This group is one member of a larger robustness family: the proof-level script independently varies the five signs of Convention 2.1(C7), the left/right placement of each of the three transport corrections, either conjugating arc for the Bs correction, and either arc, sign, and pre/post-B placement for the based T_β push-off correction. These choices give $2^5 \cdot 2^3 \cdot 2 \cdot (2 \cdot 2 \cdot 2) = 4{,}096$ relation systems. Coset enumeration terminates with order one in every case. In particular it does so for the geometrically derived member, proving the group-theoretic input to Theorem D.

For diagnostic purposes, the same formulas were evaluated at the nine formal exponent pairs $(m, n) \in \{-1, 0, 1\}^2$ and in a second based diagram. These extra computations are not used to identify the fundamental group of another manifold. Two standard procedures decide the resulting finitely presented groups; the scripts and full run records are ancillary files (Appendix A).

Coset enumeration. Six of the nine formal exponent pairs (the Lidman–Piccirillo surgery $(0, 0)$ among them) are decided outright: the enumeration terminates with $|G| = 1$, in all 32 sign assignments of Convention 2.1(C7). The $n = +1$ column resists: after Tietze simplification to 3 generators and 7 relators, those enumerations exceed 8×10^6 cosets in the full-grid runs, and 10^8 in a dedicated deeper enumeration, without terminating. Non-terminating enumeration on a presentation of the trivial group is a classical phenomenon [HEO05], and a non-terminating enumeration proves nothing in either direction; the resistant column is decided by rewriting instead.

Knuth–Bendix completion. Shortlex Knuth–Bendix completion (kbmag [KBMAG]) decides every formal exponent pair, including the resistant column, in under a second each: the completion reaches confluence, the language of irreducible words has size one, and *every generator reduces to the identity.* This last test is sufficient for triviality: the rewriting rules constructed by the program are consequences of the defining relators, and the script verifies that the normal form of every generator is the empty word. The distributed log records the aggregate pass/fail result, not a rule-by-rule completion trace; this is reproducible software verification rather than a standalone formal certificate. Controls: the same procedure run on the genus-2 surface group builds its automatic structure and reports an infinite group; run on the (infinite) group presented by the clean relations alone, it does not complete; and a run accidentally given a wrong-convention commutator (hence a different group) returns no triviality decision. An independently implemented completion program (MAF [MAF]) reproduces the conclusions on eight representative presentations. Before the resistant column was decided, a finite-quotient search had probed the two hardest formal systems from below: the groups are perfect, have no proper subgroup of index at most 7, and admit no nontrivial finite quotient of order at most 10^5: exactly what the trivial group exhibits (Appendix A).

Remark 11.1 (Robustness of the conclusion)**.** The 4,096-case check is the fixed-V robustness computation used in the proof. The larger nine-pair diagnostic grid is also 288/288 trivial. A second based diagram (a different representative of $\bar{\beta}$, whose derivation changes the Bs relation by exactly one further meridian factor, the second potential crossing vanishing because $\varphi_0^{-1}(s) \cap e = s \cap a = \varnothing$) across its full grid is likewise decided after adjoining R_3 (576/576 trivial). The pre-R_3 systems were retained as diagnostics: many coset enumerations overflow, and no conclusion is drawn from those overflows. Across the 1,408 diagnostic systems tabulated in Appendix A.2, every abelianization vanishes and no terminating enumeration produces a finite nontrivial group. The theorem, however, uses only the geometrically derived fixed-V member of the 4,096-case family. The corrected push-off word of Section 8.5 makes the presentations markedly *harder* to enumerate than the earlier, wrong word. Set beside the fact that wrong words make trivial outcomes generic (Section 6.3), this is the cautionary pairing that shaped the verification standard of this paper. Full tables: Appendix A.

Proof of Theorem D. The 4,096-case coset enumeration proves that every member of the finite robustness family is trivial, including the geometrically derived relation system. Lemma 10.2 gives an epimorphism $G \twoheadrightarrow \pi_1(V)$, whose source is trivial; hence $\pi_1(V) = 1$. □

12. Proof of Theorems A, B and C

Proof. Theorem D supplies the hypothesis $\pi_1(V) = 1$ of Theorem 1.2(c) for the specified piece. By Remark 5.1, its double also has trivial fundamental group. Part (b) then shows that Z is homeomorphic and not diffeomorphic to $S^2 \times S^2$ and that (B, W) is a homeomorphic, non-diffeomorphic pair distinguished by the sliceness of 4_1. Part (c) shows that the reglued manifold is simply connected, homeomorphic and not diffeomorphic to $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. These are Theorems A–C. □

Appendix A. Machine verification

The geometric inputs are proved in the body: the marked-surface identification in Lemma 7.1, the transport-annulus arguments of Section 8.3, the framing identification in Lemma 8.2, and the drilled-fiber basis in Lemma 10.1. The ancillary files do not prove those geometric identifications. They reproduce the algebraic developments from the stated combinatorial descriptions, the numerical run summaries, the consistency tests, and the finite-presentation decisions. Specifically, the archive contains the model and experiment scripts of Part I (`monodromy_check2.g`, `model_check3.g`, `pi1_grid.g`, `pi1_v2b.g`, `ap_check.g`); the development algorithm (`develop.py`) with its run record `develop_out.txt`, whose blocks D1–D5 are the records cited by number in Table 1; the decision scripts (`decide.g`, `fixed_v_certify.g`, `decide2.g`, `vdiag2.g`, `placement_check.g`, `vr_check.g`); the Knuth–Bendix checks of the full grid in both diagrams (`kb_certify.g`, `kb_diag2_full.g`; they require the `kbmag` package [KBMAG]) with their run records; the calibration scripts of Appendix B (`decide_t4.g`; `decide_seifert.g`, `confirm_coherent.g`); the independent MAF cross-checks (`maf_export.g`, `maf_certify.sh`, `maf_export2.g`, `maf_certify2.sh` [MAF]) with both run records; the deep-enumeration record (`tc_deep.g`); the finite-quotient scripts (`phase2_*`, `phase3_*`); and all run logs. The GitHub repository separately provides `papers/walkthrough.pdf`; that expository companion is not contained in the arXiv ancillary archive.

A.1. **The development algorithm and its validation.** A based loop transverse to the edge circles is specified by a finite combinatorial description: the departure corner wedge V_i, the ordered list of edges crossed, and the arrival wedge V_j. In the universal cover tiled by octagons, the tile with corner V_i at the base lift is $w_i^{-1}D_0$, where w_i is the length-$(i-1)$ prefix of the relator; crossing out of a tile through its edge E_i composes the deck element with $g_{\mathrm{out}}(i) = w_i\, w_{\mathrm{partner}(i)+1}^{-1}$ ($E_1{\sim}E_3$, $E_2{\sim}E_4$, $E_5{\sim}E_7$, $E_6{\sim}E_8$, reversed). The based word of such a description is $w_i^{-1}\big(\prod g_{\mathrm{out}}\big)w_j$, reduced by Dehn's algorithm for the genus-2 relator (surface groups have Dehn presentations, so the reduction decides the word problem). The algorithm (`develop.py`) implements exactly this and nothing else; every input word quoted in the body of the paper is its output on the stated description. Supplying that description is a geometric input, not a task performed by the algorithm.

Validations (all machine-checked; output in the ancillary run record).

V1. Every edge-parallel description returns its generator $(x, y, x^{-1}, r, s, s^{-1})$.

V2. The vertex-link loop (crossing all eight edges in the order $E_1E_4E_3E_2E_5E_8E_7E_6$ dictated by the link walk $V_1V_4V_3V_2V_5V_8V_7V_6$) develops to 1: a single global identity constraining all eight decks and the link structure simultaneously.

V3. The push-off descriptions return $a \simeq x^{-1}$, $b \simeq y$, $e \simeq r^{-1}$, $d \simeq s$, exactly swap-equivariantly.

V4. The two conjugating-path routes of Section 8.3 differ by a based copy of the closed curve c (the slide-around-the-curve identity) exactly.

V5. The frozen chord coordinates realize the intersection table of Figure 1, and the displayed V_7–V_3 middle-region path is strictly disjoint from both c-chords.

A.2. **The decision record.** The tables below record every decision run. Enumerations are capped at 4×10^5 cosets; "overflow" means only that the enumeration exceeded the cap. An overflow carries no group-theoretic conclusion. The Knuth–Bendix row records the R_3-augmented systems that were subsequently decided by rewriting.

run set	cases	trivial	overflow	other
proof-level fixed-V check with R_3	4,096	4,096	0	0
initial relation system (i)–(v), all (m,n) and signs	288	116	172	0
placement probe with R_3 at $(0,0)$ (2^3 placements $\times$ signs)	256	256	0	0
alternative based diagram (`vdiag2.g`)	576	8	568	0
relation system with R_3	288	192	96	0
Knuth–Bendix triviality checks (both full grids with R_3)	864	864	0	0

Every one of the 1,408 diagnostic relation systems has $H_1 = 0$. No terminating enumeration produced a finite nontrivial group. These facts are diagnostics, not decisions of the overflow cases. The corrected push-off basing word (Section 8.5) makes the presentations markedly harder for coset enumeration than their pre-repair versions (overflows are more frequent, and in the second diagram nearly universal), but the decisions are unchanged. After adjoining R_3, six of the nine formal (m,n) systems are enumeration-trivial in *all* 32 sign conventions (the Lidman–Piccirillo system $(0,0)$ among them), and the $n = +1$ column resists enumeration; the rewriting computations then decide *every* formal system in both diagrams, in every convention. The second diagram (a detour representative of $\bar{\beta}$, whose derivation reduces to a single extra meridian factor $N^{\pm}$ in the Bs relation: the second potential crossing vanishes because $\varphi_0^{-1}(s) \cap e = s \cap a = \varnothing$, and the conjugating path of M is unchanged because the detour-transport loop is trivial by the intersection data) agrees with diagram 1 wherever enumeration decides anything, and rewriting decides its full grid.

The enumeration-resistant formal systems. The pairs $(\pm 1, +1)$ resisted every coset enumeration in every version of the derivation (Tietze simplification reduces each presentation to 3 generators and 7 relators, and the enumerations exceed 8×10^6 cosets in the full-grid runs and past 10^8 in a dedicated deeper enumeration), and with the corrected push-off basing word the whole $n = +1$ column joins them. A finite-quotient search (scripts and run records in the ancillary files) probed the two hardest systems from below, in eight representative sign conventions (the two uniform and the two alternating patterns per system, in the pre-repair form of the presentations): the groups are perfect, have no proper subgroup of index at most 7 (low-index enumeration), and admit no epimorphism onto any of the 31 nonabelian simple groups of order at most 10^5; since a nontrivial finite quotient of a perfect group surjects onto a nonabelian simple group of no larger order, *no nontrivial finite quotient of order at most* 10^5 (≈ 56 CPU-hours of GAP computation). All of these are exactly what the trivial group exhibits when probed from below.

Knuth–Bendix completion (kbmag [KBMAG], shortlex, on the simplified presentations) decides every formal exponent system in under a second each, with the test structure and the three controls described in Section 11. With the corrected, sign-coupled word, the computations decide the *entire* relation-system grid (288/288 trivial, agreeing with the enumeration on all 192 cases it decided) and the *entire* second diagram (576/576: all nine exponent pairs and all 64 sign conventions each). Without R_3, Knuth–Bendix is inconclusive on diagram 2's hard formal systems: there is no confluence and therefore no decision. An independently implemented completion program, MAF [MAF], reproduces the kbmag conclusions on the eight representative presentations, run on the pre-repair word and, separately, on the corrected word: in all eight cases both times the word acceptor accepts exactly one word and every generator reduces to the identity, while the surface-group control comes out infinite (both run records in the ancillary files). Enumeration overflow past 10^8 cosets on a presentation of the trivial group is a classical phenomenon [HEO05]; it is why "overflow" was never read as "nontrivial" (the two-sided caution this pairing illustrates is drawn out in Remark 11.1).

A.3. **Verification guide.** One quantitative fact frames this section (run record in the ancillary files): perturbing the derived input words one at a time at the surgery variant (dropping the drift, inverting a push off, restoring the pre-repair word, flipping the correction sign) leaves the presented group trivial in 14 of 15 cases, with $H_1 = 0$ throughout. The decision procedures alone, in other words, say almost nothing about whether the words are geometrically correct; correctness rests on the geometric derivations and model checks below, together with the calibrations of Appendices B.1 and B.2, never on the decision outcome. The critical steps of the derivation chain, in decreasing order of delicacy, each with the machine check to consult:

(1) **The T_α framing word** $\mathrm{dir}^{\mathrm{base}}_{T_\alpha} = Ar^{-1}$ (Section 8.4). This is the Akhmedov–Park-Lemma-7-type datum. The framing identification itself is now proved (Lemma 8.2, by explicit Weinstein charts, including proofs of the two Moser normalizations it uses); the residual risk has narrowed to: the crossing-at-η conjugation formula, the identities $\varphi_0(c_y) = c_s$ and $\tilde{\varphi}(y_1) = s_1$, and the correspondence between the geometric push-off curves of the lemma and the developed words (Appendix A.1). The T^4 calibration (Appendix B.1) does *not* constrain this item (trivial monodromy has no drift analogue), but the Seifert calibration (Appendix B.2) now does: the point-push monodromy, the meridian structure, the transport law and the coherent drift words reproduce independently classified groups at three surgery coefficients, with seven wrong-word families separated. The arc pairing at the degenerate basing point c_y is also certified (the c_y-resolved test of Appendix B.2): each basing-arc side determines its transport conjugator and push-off label, the two labels one c-multiple apart; the only residual convention is the declared side, absorbed by the n-range and covered by the both-arc grid certification (Section 11).

(2) **The correction architecture** (Sections 8.3 and 8.5): the claim that basing the meridians along the partial-loop arcs y_1, s_2 makes two corrections exactly $M^{\pm}, N^{\pm}$; the conjugating path $\delta = r^{-1}$ and its V4 identity; the crossing count behind the push-off basing correction — one interior crossing of the square swept by s_2 around $\bar{\beta}$ with T_α, at (c_s, wrap), verified by the intersection data (the c-crossing precedes the e-crossing on the s-edge, itself now machine-checked: a chord-diagram computation shows this order is the unique planar realization of the model's intersection pattern, forced by $c \cap e = \varnothing$) and by $\bar{\beta}$'s single α-cut crossing — and its sign anti-coupling to the Bs-correction; the left-placement convention (probed by the 256-case placement check, but a systematic annulus-orientation error common to all placements is conceivable; such an error is now also constrained by the T^4 calibration, which it would fail). This item is exactly the error class the calibration detects, and the one place it has already found (and led to the repair of) an error.

(3) **The marked model and R_3** (Lemmas 7.1 and 10.1): the five-chain fills the genus-2 surface, the equivariant ribbon-graph identification sends the marked curves to Figure 1, and the explicit tree collapse places κ_3 in $F \smallsetminus \nu c$. Record D5 checks the frozen coordinates and the word reading, but the filling and tree-collapse arguments are geometric proofs in the text.

(4) **The model conventions** (Section 7): the cut-square gluing directions; the identification of $\bar{\alpha}, \bar{\beta}$ holonomies with $\tilde{\varphi}, \tilde{\psi}$ (an inversion here is an ε-flip, covered by Remark 11.1); the octagon link walk (validated by V2, which an erroneous link structure would fail).

(5) **The development algorithm** (Appendix A.1): validated by V1–V5; an error here would have to survive all five validations.

The rewriting-system runs add a decision procedure on top of the enumeration (`kbmag`, corroborated case-by-case by the independently implemented MAF [MAF]). For the implication used here, the script constructs the rewriting system and verifies that each generator has empty normal form. Correctness therefore relies on the implementation's construction of rules from relator consequences; the distributed logs record the outcome rather than a formal completion trace. Confluence and the language-size computation are corroborating (`kb_certify.g`, `maf_certify.sh`, and their run records, ancillary files).

A.4. **Comparing an independently derived relation sheet with ours.** Suppose an independently derived based relation sheet for $\pi_1(V)$, using the same eight generators, is at hand and does not visibly agree with Table 1. Four discrepancies carry no geometric content and should be quotiented out first:

(1) *Global inversions* of any generator: direction conventions, the ε-flips of Remark 3.7.

(2) *The five signs* $\varepsilon_3, \varepsilon_4, \varepsilon, \varepsilon_A, \varepsilon_B$ of (C7): the conclusion holds for all 32 assignments, so a sign mismatch is not a disagreement.

(3) *Basing conjugation*: any relator may be conjugated, and any meridian re-based ($\mu \mapsto w\mu w^{-1}$), without changing the presented group.

(4) *The arc identity*: $\delta = r^{-1}$ and $\delta' = x$ differ by a based copy of c (validation V4), immaterial in the group (Section 8.5); at the level of the drift word the two labels differ by one unit of n, which the n-range covers (Appendix B.2).

A relator that fails to match modulo (1)–(4) is *the* disagreement, and it is localized to a single geometric datum: one crossing count, one conjugating path, or one framing word. Each of ours carries its derivation and verification pointer in the third column of Table 1.

Two structural checks apply. An exact candidate presentation must have the known abelianization: $H_1 = 0$ after both fillings, $\mathbb{Z}$ with one filling, and $\mathbb{Z}^2$ with neither. More basically, any relation system built from a generating set and relations true in the manifold surjects onto π_1, as in Lemma 10.2. A trivial relation group therefore proves the manifold group trivial. A nontrivial relation group alone proves nothing about the manifold group; a contrary nontriviality claim also requires an independent proof that its presentation is exact.

An error found in any item above would itself be of interest: by the sensitivity analysis of Section 6.3, it would localize the exact based-loop datum on which the exotic $S^2 \times S^2$ problem turns.

Appendix B. Known-answer calibrations

The two tests summarized in Section 8.6, in full.

B.1. **The Baldridge–Kirk T^4 configuration.** The procedure that produces and decides these presentations (the cut-model conventions, the transport-annulus crossing counts, the meridian basings and conjugating paths, and the filling format) was calibrated on a configuration whose answer is known independently: the disjoint Lagrangian tori $T_1 = X \times A_1$, $T_2 = Y \times A_2$ in T^4 of Baldridge and Kirk [BK09], whose complement presentation they computed with explicit care about basings, and for which double ± 1 Luttinger surgery gives $\pi_1 = (\mathbb{Z}^2 \rtimes_A \mathbb{Z}) \times \mathbb{Z}$ with $|\operatorname{tr} A| \in \{1, 3\}$ according to the relative signs, non-abelian in every convention. Deriving the input words for this configuration by the same transport-annulus method and checking all 64 sign/placement conventions: with the derived conjugating-path data (the pair $(1, b)$, together with its $T_1 \leftrightarrow T_2$ mirror $(b, 1)$, an exact symmetry of the configuration), the decision scripts reproduce the known answer in all 64 cases each, with the exact $\operatorname{tr} = 3/\operatorname{tr} = 1$ split; all seven other pairs (including the chirality flip) fail all 64, six of them with half their failures abelian, the signature of a missed basing-arc crossing, and the double-flip (b^{-1}, b^{-1}) with wrong non-abelian groups throughout (per-pair counts in the run record). A single surgery reproduces $\mathcal{H} \times \mathbb{Z}$ ($\mathcal{H}$ the integer Heisenberg group) in all conventions, and Baldridge–Kirk's published words, run through the same decision procedure, match ours case by case. The test detects single basing-arc errors, and it is what found the correction of Section 8.5: the first, hand-derived T^4 words failed exactly as the test's acceptance criteria (fixed in advance) describe, the diff against the published presentation localized the error to one missed circle-slide crossing, and re-examining this derivation for the same kind of error found it in — and only in — $\operatorname{dir}^{\text{base}}_{T_\beta}$ (the symmetric candidates come out clean: $y_1 \cap e = \varnothing$ protects $\operatorname{dir}^{\text{base}}_{T_\alpha}$, and $\operatorname{dir}^{\text{fib}}_{T_\beta}$'s basing-arc tail is crossing-free). The scripts, the design (fixed in advance of the runs) and the run record are in the ancillary files. One scope note: T^4 has trivial monodromy, so this calibration does not exercise the monodromy-drift machinery behind $\operatorname{dir}^{\text{base}}_{T_\alpha} = Ar^{-1}$; that machinery has its own known-answer test (Appendix B.2), with the framing side in Lemma 8.2.

B.2. **A Seifert-fibered test of the base-direction push off.** A second known-answer configuration exercises exactly the machinery the T^4 test cannot: the same fiber F, the same φ_0, the same invariant curve c, but trivial vertical monodromy: the closed manifold $X = N \times S^1$ with $N = \mathrm{MT}(\varphi_0)$, Seifert fibered over the torus with two cone points of order 2, surgered once along $T_\lambda = \lambda \times S^1$, the drift section itself times the extra circle, with the T_α-framing. Here the drift cannot hide: λ meets every fiber of N exactly once, so the drilled complement fibers with *once-punctured* fiber and monodromy equal to the swap composed with the point-push along the drift arc, and the meridian is a conjugate of the boundary word $[x, y][r, s]$ rather than a free generator, which turns the meridian-transport law from an input into a machine-checkable consequence of the other input words. The known answer is classical: $1/k$-surgery on a curve lying in an incompressible torus, taken with the torus framing, is a k-fold torus twist, so the surgered manifolds are graph manifolds with one JSJ wall, presented independently over the Seifert piece of the wall complement. The known-answer family is anchored without reference to the algorithm: its untwisted closure must reproduce $\pi_1(N)$ (it does, on the computed invariants; a deliberately wrong wall word fails with exactly the predicted homology defect), and $H_1 = \mathbb{Z}^2 \oplus \mathbb{Z}/|k|$, computed by hand on both sides, must match (it does).

The outcome, from the design fixed in advance (scripts and run records in the ancillary files): the coherence constraints pin the package of input words (meridian basing, transport conjugator, push-off label) essentially uniquely, and it is the derived one, with the transport conjugator literally the conjugating path $\delta = r^{-1}$ and the meridian sign anti-coupled to the sign of the transport relation's meridian factor, the same anti-coupling as in Section 8.5; the second power of the monodromy carries the meridian to its (rx)-conjugate (the drilled form of the drift identity λ^2) on the nose; and with the coherent words the decision scripts reproduce the classified groups at $k \in \{1, -1, 2\}$ in both mirror-image packages. Seven wrong-word families (reversed drift, ordering flip, the basing-conjugate family that every earlier test left undecided, a commutator in place of the meridian factor, a dropped point-push, an arc-incoherent meridian factor, and a meridian slip in the framing) are separated from the known answer at every coefficient: the discriminating power the decision procedures alone lack (Appendix A.3).

The calibration also caught one thing, which we disclose in the same spirit as Section 8.5: the word Ar^{-1}, transplanted verbatim from the c_y-based display of Section 8.4 into the test's differently based filling relation, matches nothing, while the coherent word at that basing is the other-arc representative $Ax = Ar^{-1} \cdot (rx)$. At word level, then, the arc pairing is basing-dependent within the $\lambda + \mathbb{Z}c$ family of drift sections (exactly the one-unit-of-n ambiguity already noted after the display of Section 8.4), and the test certifies the machinery (monodromy, meridian structure, transport law, and the coherence rules tying basing arc, arc route and sign together).

A follow-up test resolves the degenerate basing point itself: puncture the fiber *at* c_y and split the y-generator at the puncture, so the side-of-c choice becomes a symbol in the presentation. The outcome is sharper than a dictionary of conventions. No meridian-free package is correct (the meridian-free form satisfies the transport law with longer conjugators yet fails the known answer at every coefficient, checked directly, not merely pruned), and the basing arc's approach side determines the entire package: the y_1-side meridian basing is coherent with transport conjugator r^{-1} (literally the corr_{Bs} conjugating path of the Bs correction: for the meridian *transport*, the drift is exactly that path) and push-off label Ax, while the y_2-side is coherent with x-transport and push-off label Ar^{-1}; all cross-pairings fail. The display's word Ar^{-1} is thus the y_2-side entry of a certified two-entry dictionary whose entries differ by that c-multiple; the only residual convention is which side one declares, and no conclusion is sensitive to it: the n-range covers the shift, and the grid certification of Section 11 was run with both arcs.

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